

Semi-Implicit and Unconditionally Energy-Stable Finite Element Schemes for the Rosensweig Model for Ferrofluid Flows*

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Abstract

We develop a unified structure-preserving framework for the numerical approximation of the Rosensweig model for ferrofluid flows and construct first- and second-order semi-implicit finite element schemes that are unconditionally energy stable. By introducing a scalar magnetic potential and suitable auxiliary variables, the strongly coupled system is reformulated as an abstract saddle-point problem in which the dissipative terms are represented by a symmetric coercive bilinear form and the nonlinear couplings by a skew-symmetric trilinear form. This formulation exposes the cancellation mechanism of the coupled system and provides a common framework for designing energy-stable schemes of different temporal orders. The first-order method is based on a modified implicit Euler discretization and is linear at every time level, while the second-order method combines a modified Crank-Nicolson discretization with extrapolation and requires a nonlinear solve only at the initial step. For the spatial discretization, compatible H^1 -, $H(\text{curl})$ -, and $H(\text{div})$ -conforming finite element spaces are employed within the framework of finite element exterior calculus. In particular, the discrete magnetic field is recovered as the gradient of a discrete scalar potential and therefore satisfies the curl-free constraint pointwise. We establish well-posedness and unconditional energy estimates for the time semi-discrete and fully discrete schemes, together with first- and second-order temporal error estimates. An exact Schur complement decoupling strategy is developed for the resulting linear algebraic systems. Numerical experiments for both schemes confirm the expected convergence behavior and the preservation of the discrete curl-free property up to machine precision. Additional straight-channel and backward-facing-step examples illustrate magnetically forced transport under prescribed inflow and open-boundary conditions.

Keywords: *ferrofluid; semi-implicit; unconditional energy stability; structure-preserving; error estimates.*

1 Introduction

Ferrofluids, also referred to as magnetic fluids, are stable colloidal suspensions of nanoscale ferromagnetic or ferrimagnetic particles in a carrier liquid. Owing to their strong response to externally applied magnetic fields, ferrofluids combine the flow characteristics of conventional liquids with magnetically controllable behavior [28, 29, 31]. This distinctive feature has led to a broad range of applications in engineering and technology, including magnetic sealing, lubrication, damping and vibration control, heat-transfer enhancement, sensing and actuation, as well as microfluidic and other magnetically controlled devices. Ferrofluids have also attracted considerable attention in biomedical applications, such as magnetic drug targeting, magnetic hyperthermia, and magnetic-resonance-related technologies [18]. These applications require an accurate description of the interaction among fluid motion, particle rotation, magnetization, and the applied magnetic field, and consequently motivate the development of reliable and efficient numerical methods for ferrohydrodynamic systems.

Several continuum models have been proposed to describe ferrofluid flows. Among them, the models of Shliomis [30, 31] and Rosensweig [28, 29] are two of the most widely studied. The Rosensweig model considered in this work consists of the incompressible Navier-Stokes equations for the macroscopic fluid velocity, an angular-momentum equation describing particle spin, a transport equation for the magnetization, and the magnetostatic equations. Compared with the simplified Shliomis-type models, the additional angular momentum equation introduces further linear and nonlinear couplings between the fluid velocity, particle rotation, magnetization, and magnetic field. The resulting system is therefore strongly coupled and highly nonlinear and, at the same time, possesses important physical structures, including incompressibility, magnetostatic constraints, and an energy-dissipation law. Analytical properties of ferrohydrodynamic models, including the existence of global weak solutions, strong solutions, and uniqueness under suitable assumptions, have been investigated in [1, 2, 3, 4]. For the Rosensweig system, Nohetto, Salgado, and Tomas further studied its mathematical structure, energy law, and long-time behavior [24, 25].

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The strong nonlinear coupling and the coexistence of several differential structures make the numerical approximation of ferrohydrodynamic systems particularly challenging. In addition to accuracy, a desirable numerical method should reproduce, at the discrete level, fundamental structural properties of the continuous model, such as the energy-dissipation law and the differential constraints imposed on the magnetic variables. Early numerical studies demonstrated the effectiveness of finite element and related discretizations for magnetically driven ferrofluid flows. A particularly important development was made by Nochetto, Salgado, and Tomas [24], who formulated the Rosensweig system in an energy-stable framework and constructed a numerical scheme that mimics its continuous energy estimate. They also developed an energy-stable diffuse-interface formulation for two-phase ferrofluid flows in [23]. These studies demonstrated that preserving the intrinsic energy structure is not only physically meaningful but also provides an effective tool for establishing robust stability of numerical approximations.

Finite element methods are particularly attractive for ferrohydrodynamic problems because the different physical variables naturally belong to different function spaces. Standard H^1 -conforming elements are suitable for the fluid velocity and angular velocity, while $H(\text{curl})$ - and $H(\text{div})$ -conforming elements provide natural approximations for magnetic variables. In this respect, finite element exterior calculus provides a systematic framework for constructing compatible spaces and discrete differential operators through the discrete de Rham complex [6, 7, 12]. Such compatible discretizations have proved particularly successful in magnetohydrodynamics, where exact preservation of divergence or curl constraints is essential for stability and physical fidelity [14, 15, 13, 26]. These ideas are equally relevant to ferrohydrodynamics, since the magnetostatic equation imposes a curl-free condition on the magnetic field.

In recent years, considerable progress has been made in the finite element approximation of ferrohydrodynamic systems. For Shliomis-type models, Wu and Xie developed energy-stable mixed finite element methods and established stability, well-posedness, and error estimates [32]; mixed finite element approximations for a ferrofluid model with magnetization parallel to the magnetic field were further studied in [33]. Mao and Sun [20] analyzed a first-order Euler semi-implicit mixed finite element method for the incompressible Shliomis model and established unconditional error estimates. Second-order, linearized, and unconditionally stable schemes for ferrohydrodynamic systems have also been investigated; see, for example, Keram and Huang [16]. These developments indicate the importance of combining temporal linearization with an appropriate treatment of the underlying energy structure.

More recently, numerical approximation of the full Rosensweig model has received increasing attention. Dong, Huang, Huang, and Tang [10] proposed a linear semi-implicit, energy-stable fully discrete finite element method and established existence, uniqueness, and unconditional convergence of the numerical solutions. Wu and Xie [34] developed energy-stable mixed finite element semi-discretizations and fully discrete schemes for the Rosensweig model, proved well-posedness and optimal error estimates, and designed a decoupled quasi-Newton iteration for the nonlinear fully discrete system. Zhang, He, and Yang [35] constructed an efficient, fully decoupled, linear, second-order, and unconditionally energy-stable finite element scheme by introducing a nonlocal auxiliary variable and combining it with a zero-energy-contribution decoupling strategy. Very recently, Dong, Huang, Huang, and Yang [11] proposed a linear fully discrete finite element method with second-order temporal accuracy that preserves the original energy of the Rosensweig model. These results represent substantial progress toward efficient and energy-stable simulation of the full ferrohydrodynamic system.

Among these studies, the mixed finite element framework of Wu and Xie [34] is the most closely related to the present work. In [34], the magnetic field is treated as an independent $H(\text{div})$ -conforming unknown in a mixed formulation, and the fully discrete scheme is nonlinear and is solved by a decoupled quasi-Newton iteration; optimal error estimates are established for the corresponding semi-discrete and fully discrete approximations. The present work takes a different route. By introducing a scalar magnetic potential together with suitable auxiliary variables, we recast the coupled equations into an abstract saddle-point system whose dissipative and nonlinear coupling parts are represented, respectively, by symmetric and skew-symmetric forms. This structure yields both first- and second-order semi-implicit schemes within the same framework. Moreover, the magnetic field is recovered as $\mathbf{H}_h = \nabla \varphi_h$, so that the discrete curl-free constraint is satisfied pointwise. The first-order method is linear at every time level, while the second-order method is linear after the starting step, and the resulting linear systems admit the same exact Schur complement decoupling strategy. In this sense, the present study complements and extends the earlier mixed finite element analysis in [34] from the viewpoint of a unified structure-preserving formulation, higher-order temporal discretization, exact magnetic constraint preservation, and block algebraic solution.

Nevertheless, it remains desirable to develop a formulation that exposes the cancellation mechanism of the coupled equations at an abstract level, so that energy-stable schemes of different temporal orders can be constructed within a common framework. It is also important to combine this temporal structure with compatible finite element spaces that retain the differential structure of the magnetic field and with solvers that exploit the resulting block algebraic structure.

The purpose of this paper is to develop such a unified structure-preserving framework for the Rosensweig ferrofluid flow model. The main idea is to reformulate the original system so that its dissipative and conservative coupling mechanisms become transparent. More precisely, by introducing a scalar magnetic potential together with suitable auxiliary variables, we rewrite the governing equations as an abstract saddle-point problem of the form

$$\begin{cases} \mathcal{D}(\partial_t \mathcal{X}, \mathcal{Y}) + \mathbf{A}(\mathcal{X}, \mathcal{Y}) + \mathbf{B}(\mathcal{X}, \mathcal{X}, \mathcal{Y}) + \mathcal{C}(\mathcal{Y}, \tilde{p}) = \langle \mathbf{f}, \mathcal{Y} \rangle, \\ \mathcal{C}(\mathcal{X}, q) = 0, \end{cases}$$

where $\mathbf{A}(\cdot, \cdot)$ is symmetric and coercive and $\mathbf{B}(\cdot, \cdot, \cdot)$ is skew-symmetric with respect to its second and third arguments. In

particular,

$$\mathcal{B}(\mathcal{Z}, \mathcal{Y}, \mathcal{Y}) = 0,$$

which provides a unified algebraic mechanism for reproducing the continuous energy-dissipation law at the discrete level.

The main contributions of this work can be summarized as follows.

- (i) We derive a new abstract formulation of the Rosensweig system in which the symmetric dissipative terms and the skew-symmetric nonlinear couplings are separated explicitly. Unlike a discretization constructed term by term directly from the original equations, this formulation provides a unified framework for the systematic design and analysis of energy-stable schemes of different temporal orders.
- (ii) Based on this abstract structure, we construct both first- and second-order semi-implicit time discretizations. The first-order method is based on a modified implicit Euler scheme, whereas the second-order method combines a modified Crank-Nicolson discretization with extrapolation of the nonlinear coefficient. Both methods are unconditionally energy stable. The first-order scheme is linear at every time level. For the second-order method, only the starting step requires the solution of a nonlinear problem, while all subsequent time levels involve linear systems. Moreover, in the absence of an external magnetic field, the second-order scheme satisfies the corresponding discrete energy-dissipation identity.
- (iii) For the spatial approximation, we employ compatible H^1 -, $H(\text{curl})$ -, and $H(\text{div})$ -conforming finite element spaces within the framework of finite element exterior calculus. The associated discrete weak curl and discrete Laplace operators are constructed consistently with the continuous differential structure. In particular, the discrete magnetic field is recovered as

$$\mathbf{H}_h = \nabla \varphi_h,$$

and therefore

$$\text{curl } \mathbf{H}_h = 0$$

holds pointwise. Thus, the magnetostatic curl-free constraint is preserved exactly at the discrete level rather than only approximately.

- (iv) We establish well-posedness and unconditional energy estimates for the time semi-discrete and fully discrete schemes and derive first- and second-order temporal error estimates for the corresponding time semi-discrete methods. For the spatial discretization, compatible finite element spaces are employed to retain the underlying differential structure of the magnetic variables. In addition, the block structure of the fully discrete systems is analyzed and an exact Schur complement decoupling solver is developed. The same algebraic solution strategy applies to the first-order method and to the linear systems arising from the second-order method after the starting step. Numerical experiments for both the first- and second-order methods illustrate the convergence behavior of the fully discrete schemes and demonstrate that the discrete curl-free constraint is preserved up to machine precision.

The remainder of the paper is organized as follows. In Section 2, we introduce the functional setting, the notation for time discretization, the compatible finite element spaces, and the associated discrete differential operators. Section 3 presents the Rosensweig ferrofluid flow model, derives its structure-preserving weak formulation, and introduces the abstract formulation used throughout the paper. In Section 4, we develop the first-order semi-implicit scheme, prove its well-posedness and energy stability, derive temporal error estimates, introduce the compatible fully discrete finite element formulation and its spatial consistency analysis, and present the exact Schur complement decoupling solver. Section 5 is devoted to the second-order semi-implicit scheme based on a modified Crank-Nicolson discretization. Its well-posedness, unconditional energy stability, temporal error estimates, compatible fully discrete finite element formulation, and algebraic solution strategy are discussed. Section 6 collects the convergence and unforced energy-dissipation experiments for both time discretizations and presents straight-channel and backward-facing flow applications.

2 Preliminaries

In this section, we establish the functional and discrete settings required for the subsequent formulation and analysis. We first introduce the relevant Sobolev spaces, weak differential operators, and product Hilbert spaces associated with the ferrohydrodynamic system. We then specify the notation for time discretization and collect several consistency identities that will be used in the temporal error analysis. Finally, we introduce the conforming finite element spaces for the spatial discretization, together with their inf-sup stability and commuting exact-sequence properties, and define the corresponding discrete differential operators.

2.1 Sobolev spaces

Let $\Omega \subset \mathbb{R}^3$ be a bounded, simply connected, and convex domain with Lipschitz boundary $\partial\Omega$, and let $T > 0$ denote the final time. We set

$$\Omega_T := \Omega \times (0, T], \quad \Gamma_T := \partial\Omega \times (0, T],$$

and denote by $\mathbf{n}$ the unit outward normal vector on $\partial\Omega$. For any nonnegative integer l , let $H^l(\Omega)$ denote the standard Sobolev space of order l , equipped with the norm $\|\cdot\|$ and seminorm $|\cdot|_l$. For the corresponding vector-valued spaces $(H^l(\Omega))^3$ and $(L^2(\Omega))^3$, we use the same notation for norms, seminorms, and inner products as in the scalar case.

We further introduce the spaces

$$H(\text{curl}) := \{\mathbf{v} = (v_1, v_2, v_3)^\top \in (L^2(\Omega))^3 : \text{curl } \mathbf{v} \in (L^2(\Omega))^3\},$$

where

$$\text{curl } \mathbf{v} = (\partial_y v_3 - \partial_z v_2, \partial_z v_1 - \partial_x v_3, \partial_x v_2 - \partial_y v_1)^\top,$$

and

$$H(\text{div}) := \{\mathbf{v} = (v_1, v_2, v_3)^\top \in (L^2(\Omega))^3 : \text{div } \mathbf{v} \in L^2(\Omega)\},$$

where

$$\text{div } \mathbf{v} = \partial_x v_1 + \partial_y v_2 + \partial_z v_3.$$

These spaces are equipped with the norms

$$\|\mathbf{v}\|_c = (\|\mathbf{v}\|^2 + \|\text{curl } \mathbf{v}\|^2)^{1/2}, \quad \|\mathbf{v}\|_d = (\|\mathbf{v}\|^2 + \|\text{div } \mathbf{v}\|^2)^{1/2}.$$

We define the following function spaces:

$$\mathbf{S} := (H_0^1(\Omega))^3 = \{\mathbf{v} \in (H^1(\Omega))^3 : \mathbf{v} = 0 \text{ on } \partial\Omega\},$$

$$\Sigma := H_0^1(\Omega) = \{v \in H^1(\Omega) : v = 0 \text{ on } \partial\Omega\},$$

$$\mathbf{U} := H_0(\text{curl}) = \{\mathbf{v} \in H(\text{curl}) : \mathbf{v} \times \mathbf{n} = 0 \text{ on } \partial\Omega\},$$

$$H(\text{div } 0) := \{\mathbf{v} \in H(\text{div}) : \text{div } \mathbf{v} = 0\},$$

$$\mathbf{V} := H_0(\text{div}) = \{\mathbf{v} \in H(\text{div}) : \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \partial\Omega\},$$

$$H_0(\text{div } 0) := H_0(\text{div}) \cap H(\text{div } 0),$$

$$W := L_0^2(\Omega) = \left\{ v \in L^2(\Omega) : \int_{\Omega} v \, dx = 0 \right\}.$$

$$\mathbf{L}_n^2(\Omega) := \{\mathbf{v} \in (L^2(\Omega))^3 : \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \partial\Omega\},$$

$$\mathbf{R}^s := \{\mathbf{v} \in \mathbf{S} : \text{div } \mathbf{v} = 0\}.$$

For vector fields $\mathbf{v} \in H_0(\text{curl}) \cap H(\text{div})$ or $\mathbf{v} \in H(\text{curl}) \cap H_0(\text{div})$, we define

$$\|\mathbf{v}\|_n = (\|\text{curl } \mathbf{v}\|^2 + \|\text{div } \mathbf{v}\|^2)^{1/2}.$$

We next introduce the product Hilbert space

$$\mathbb{V} = \mathbf{S} \times \mathbf{S} \times (\mathbf{V} \cap H(\text{curl})) \times D(-\Delta), \quad \text{with } D(-\Delta) := \{\psi \in \Sigma : -\Delta\psi \in L^2(\Omega)\}$$

equipped with the inner product

$$(\mathcal{X}, \mathcal{Y})_{\mathbb{V}} = (\nabla \mathbf{u}, \nabla \mathbf{v}) + (\nabla \boldsymbol{\omega}, \nabla \mathbf{s}) + (\mathbf{m}, \mathbf{F}) + (\text{div } \mathbf{m}, \text{div } \mathbf{F}) + (\text{curl } \mathbf{m}, \text{curl } \mathbf{F}) + (\nabla \varphi, \nabla \psi) + (\Delta \varphi, \Delta \psi),$$

for any

$$\mathcal{X} = (\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \varphi)^\top \in \mathbb{V}, \quad \mathcal{Y} = (\mathbf{v}, \mathbf{s}, \mathbf{F}, \psi)^\top \in \mathbb{V}.$$

The associated norm is defined by

$$\|\mathcal{Y}\|_{\mathbb{V}} := (\mathcal{Y}, \mathcal{Y})_{\mathbb{V}}^{1/2}, \quad \forall \mathcal{Y} \in \mathbb{V}.$$

2.2 Notation for time discretization

Let N be a positive integer and set

$$\Delta t := \frac{T}{N}.$$

We consider the uniform partition

$$\mathcal{T}_{\Delta t} = \{t_n : t_n = n\Delta t, 0 \leq n \leq N\}$$

of the time interval $[0, T]$. For $1 \leq n \leq N$, let

$$I_n = (t_{n-1}, t_n),$$

and denote by v^n an approximation to $v(t_n)$.

We introduce the following difference, averaging, and extrapolation operators:

$$\begin{aligned} \delta_t v(t_n) &:= \frac{v(t_n) - v(t_{n-1})}{\Delta t}, & \bar{v}(t_{n-1/2}) &:= \frac{v(t_n) + v(t_{n-1})}{2}, \\ \delta_t v^n &:= \frac{v^n - v^{n-1}}{\Delta t}, & \bar{v}^{n-1/2} &:= \frac{v^n + v^{n-1}}{2}, \\ \hat{v}(t_n) &:= \frac{3}{2}v(t_n) - \frac{1}{2}v(t_{n-1}), & \hat{v}^n &:= \frac{3}{2}v^n - \frac{1}{2}v^{n-1}, \quad 1 \leq n \leq N, \\ \hat{v}(0) &:= \frac{v(t_1) + v(t_0)}{2}, & \hat{v}^0 &:= \frac{v^1 + v^0}{2}. \end{aligned}$$

The following identities [19, Lemmas 2.3 and 2.4] will be repeatedly used to estimate the temporal consistency errors.

Lemma 2.1 ([19]). For $1 \leq n \leq N$, let $v \in W^{2,1}(0, T)$. Then

$$\delta_t v(t_n) - \partial_t v(t_n) = \frac{1}{\Delta t} \int_{t_{n-1}}^{t_n} \partial_{tt} v(s)(t_{n-1} - s) ds.$$

Furthermore, if $v \in W^{3,1}(0, T)$, then

$$v(t_n) - v(t_{n-1}) - \Delta t \partial_t v(t_{n-1/2}) = -\frac{1}{2} \int_{t_{n-1}}^{t_n} (s - t_{n-1})(t_n - s) \partial_{ttt} v(s) ds + \frac{\Delta t}{2} \int_{t_{n-1/2}}^{t_n} \int_{s-\Delta t/2}^s \partial_{ttt} v(l) dl ds,$$

and

$$\bar{v}(t_{n-1/2}) - v(t_{n-1/2}) = \frac{1}{2} \int_{t_{n-1}}^{t_{n-1/2}} (t_{n-1} - s) \int_s^{s+\Delta t/2} \partial_{ttt} v(l) dl ds + \frac{\Delta t}{4} \int_{t_{n-1}}^{t_{n-1/2}} \partial_{tt} v(s + \Delta t/2) ds.$$

Moreover,

$$\hat{v}(t_{n-1}) - v(t_{n-1/2}) = \begin{cases} -\frac{1}{2} \int_{t_{n-1}}^{t_n} \int_{s-\Delta t}^{s-\Delta t/2} \partial_{tt} v(l) dl ds - \frac{1}{2} \int_{t_{n-1}}^{t_{n-1/2}} \int_{s-\Delta t/2}^s \partial_{tt} v(l) dl ds, & n \geq 2, \\ \frac{1}{2} \int_{t_0}^{t_{1/2}} (t_0 - s) \int_s^{s+\Delta t/2} \partial_{ttt} v(l) dl ds + \frac{\Delta t}{4} \int_{t_0}^{t_{1/2}} \partial_{tt} v(s + \Delta t/2) ds, & n = 1. \end{cases}$$

2.3 Finite element spaces

Let $\mathcal{T}_h$ be a quasi-uniform and shape-regular triangulation of Ω with mesh size $h > 0$. We introduce below the conforming finite element spaces used for the spatial discretization, including H^1 -, $H(\text{curl})$ -, and $H(\text{div})$ -conforming elements. For each $K \in \mathcal{T}_h$ and nonnegative integer l , let $\mathbb{P}_l(K)$ denote the space of polynomials on K of total degree at most l .

We employ the following conforming finite element spaces:

- $\mathbf{S}_h^i = (\mathbf{S}_h^i)^3 \subset \mathbf{S}$, $i = 1, 2$. The space $\mathbf{S}_h^1$ is the continuous piecewise polynomial space of the mini element [5], whose local space is given by

$$\mathbf{S}_h^1|_K = \mathbb{P}_1(K) + B_K, \quad B_K = \text{span} \left\{ \prod_{j=1}^4 \lambda_j^K \right\},$$

where λ_j^K , $j = 1, \dots, 4$, are the barycentric coordinates of K , and B_K is the space generated by the quartic bubble function. The space $\mathbf{S}_h^2$ is the standard continuous piecewise quadratic finite element space with

$$\mathbf{S}_h^2|_K = \mathbb{P}_2(K).$$

- $L_h \subset H^1(\Omega) \cap W$ is the standard continuous piecewise linear finite element space [9].
- $\Sigma_h^i = (\Sigma_h^i)^3 \subset \mathbf{S}$, $i = 1, 2$, where Σ_h^1 and Σ_h^2 are, respectively, the continuous piecewise linear and continuous piecewise quadratic finite element spaces.
- $\mathbf{U}_h^i \subset \mathbf{U}$, $i = 1, 2$. Here, $\mathbf{U}_h^1$ is the lowest-order Nédélec edge-element space (NE_0) [21, 22], satisfying

$$(\mathbb{P}_0(K))^3 \subset \mathbf{U}_h^1|_K,$$

whereas $\mathbf{U}_h^2$ is the first-order Nédélec edge-element space (NE_1), satisfying

$$(\mathbb{P}_1(K))^3 \subset \mathbf{U}_h^2|_K.$$

- $\mathbf{V}_h^i \subset \mathbf{V}$, $i = 1, 2$. Here, $\mathbf{V}_h^1$ is the lowest-order Raviart–Thomas face-element space (RT_0) [27], satisfying

$$(\mathbb{P}_0(K))^3 \subset \mathbf{V}_h^1|_K,$$

whereas $\mathbf{V}_h^2$ is the first-order Raviart–Thomas face-element space (RT_1), satisfying

$$(\mathbb{P}_1(K))^3 \subset \mathbf{V}_h^2|_K.$$

- $W_h^i \subset W$, $i = 1, 2$, where W_h^1 is the space of piecewise constants and W_h^2 is the space of piecewise linear polynomials [9].

The above finite element spaces satisfy the following two fundamental compatibility properties, which play a crucial role in the stability and subsequent error analysis.

(1) **Discrete inf-sup stability** [5]:

$$\sup_{\mathbf{v}_h \in \mathbf{S}_h^i \setminus 0} \frac{(q_h, \operatorname{div} \mathbf{v}_h)}{\|\mathbf{v}_h\|_1} \gtrsim \|q_h\|, \quad \forall q_h \in L_h, \quad i = 1, 2.$$

(2) **Commuting exact sequence** [12]:

$$\begin{array}{ccccccc} H_0^1(\Omega) & \xrightarrow{\operatorname{grad}} & \mathbf{U} & \xrightarrow{\operatorname{curl}} & \mathbf{V} & \xrightarrow{\operatorname{div}} & W \\ \downarrow \pi_h^i & & \downarrow \pi_h^{ci} & & \downarrow \pi_h^{di} & & \downarrow Q_h^i \\ \Sigma_h^i & \xrightarrow{\operatorname{grad}} & \mathbf{U}_h^i & \xrightarrow{\operatorname{curl}} & \mathbf{V}_h^i & \xrightarrow{\operatorname{div}} & W_h^i \end{array} \quad i = 1, 2. \quad (1)$$

Here,

$$\pi_h^i : H_0^1(\Omega) \rightarrow \Sigma_h^i, \quad \pi_h^{ci} : \mathbf{U} \rightarrow \mathbf{U}_h^i, \quad \pi_h^{di} : \mathbf{V} \rightarrow \mathbf{V}_h^i$$

denote the canonical interpolation operators, while

$$Q_h^i : W \rightarrow W_h^i$$

denotes the L^2 -orthogonal projection.

To mimic the corresponding weak differential operators at the discrete level, we define

$$\operatorname{curl}_h^i : \mathbf{V}_h^i \rightarrow \mathbf{U}_h^i \quad \text{and} \quad -\Delta_h^i : \Sigma_h^i \rightarrow \Sigma_h^i \quad i = 1, 2$$

as follows:

- For any $\mathbf{v}_h \in \mathbf{V}_h^i$, the discrete weak curl operator $\operatorname{curl}_h^i \mathbf{v}_h \in \mathbf{U}_h^i$ is defined by

$$(\operatorname{curl}_h^i \mathbf{v}_h, \mathbf{w}_h) = (\mathbf{v}_h, \operatorname{curl} \mathbf{w}_h), \quad \forall \mathbf{w}_h \in \mathbf{U}_h^i.$$

- For any $\phi_h \in \Sigma_h^i$, the discrete negative Laplacian $-\Delta_h^i \phi_h \in \Sigma_h^i$ is defined by

$$(-\Delta_h^i \phi_h, \psi_h) = (\nabla \phi_h, \nabla \psi_h), \quad \forall \psi_h \in \Sigma_h^i.$$

Corresponding to the continuous product space $\mathbb{V}$, we introduce the discrete spaces

$$\mathbb{V}_h^i = \mathbf{S}_h^i \times \Sigma_h^i \times \mathbf{V}_h^i \times \Sigma_h^i, \quad i = 1, 2.$$

For

$$\mathcal{X}_h = (\mathbf{u}_h, \boldsymbol{\omega}_h, \mathbf{m}_h, \varphi_h)^\top \in \mathbb{V}_h^i, \quad \mathcal{Y}_h = (\mathbf{v}_h, \mathbf{s}_h, \mathbf{F}_h, \psi_h)^\top \in \mathbb{V}_h^i,$$

we define

$$\begin{aligned} (\mathcal{X}_h, \mathcal{Y}_h)_{\mathbb{V}_h^i} &:= (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) + (\nabla \boldsymbol{\omega}_h, \nabla \mathbf{s}_h) + (\mathbf{m}_h, \mathbf{F}_h) + (\operatorname{div} \mathbf{m}_h, \operatorname{div} \mathbf{F}_h) \\ &\quad + (\operatorname{curl}_h^i \mathbf{m}_h, \operatorname{curl}_h^i \mathbf{F}_h) + (\nabla \varphi_h, \nabla \psi_h) + ((-\Delta_h^i) \varphi_h, (-\Delta_h^i) \psi_h). \end{aligned}$$

The associated norm is

$$\|\mathcal{Y}_h\|_{\mathbb{V}_h^i} := (\mathcal{Y}_h, \mathcal{Y}_h)_{\mathbb{V}_h^i}^{1/2}, \quad \forall \mathcal{Y}_h \in \mathbb{V}_h^i.$$

3 The Rosensweig model for ferrofluid flows

We consider an incompressible, viscous Newtonian ferrofluid occupying the domain Ω and subject to an externally applied magnetic field $\mathbf{H}_e$ satisfying the boundary condition

$$\mathbf{H}_e \cdot \mathbf{n} = 0 \quad \text{on } \Gamma_T.$$

The external field $\mathbf{H}_e$ induces a demagnetizing field $\mathbf{H}$ and a magnetic induction $\mathbf{B}$, which are related by

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{m}),$$

where $\mathbf{m}$ denotes the magnetization field.

Following the Rosensweig model [28, 29], the dynamics of the ferrofluid are governed by the conservation of linear and angular momentum, the incompressibility constraint, the magnetization transport equation, and the magnetostatic equations. More precisely, the velocity $\mathbf{u}$, pressure p , angular velocity $\boldsymbol{\omega}$, magnetization $\mathbf{m}$, and magnetic field $\mathbf{H}$ satisfy

$$\rho(\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}) - (\eta + \zeta) \Delta \mathbf{u} + \nabla p = \mu_0(\mathbf{m} \cdot \nabla) \mathbf{H} + 2\zeta \operatorname{curl} \boldsymbol{\omega} \quad \text{in } \Omega_T, \quad (2a)$$

$$\operatorname{div} \mathbf{u} = 0 \quad \text{in } \Omega_T, \quad (2b)$$

$$\rho\kappa(\partial_t \boldsymbol{\omega} + (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}) - \eta' \Delta \boldsymbol{\omega} - (\eta' + \lambda') \nabla \operatorname{div} \boldsymbol{\omega} = \mu_0 \mathbf{m} \times \mathbf{H} + 2\zeta(\operatorname{curl} \mathbf{u} - 2\boldsymbol{\omega}) \quad \text{in } \Omega_T, \quad (2c)$$

$$\partial_t \mathbf{m} + (\mathbf{u} \cdot \nabla) \mathbf{m} - \sigma \Delta \mathbf{m} = \boldsymbol{\omega} \times \mathbf{m} - \frac{1}{\tau}(\mathbf{m} - \chi_0 \mathbf{H}) \quad \text{in } \Omega_T, \quad (2d)$$

$$\operatorname{curl} \mathbf{H} = 0 \quad \text{in } \Omega_T, \quad (2e)$$

$$\mu_0 \operatorname{div}(\mathbf{H} + \mathbf{m}) = -\operatorname{div} \mathbf{H}_e \quad \text{in } \Omega_T. \quad (2f)$$

Here, the positive constants $\rho, \eta, \zeta, \mu_0, \kappa, \eta', \lambda', \tau$, and χ_0 are physical parameters; their detailed physical interpretations can be found in [30, 28, 29, 31]. The system (2) is supplemented with the initial conditions

$$\mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x}), \quad \boldsymbol{\omega}(\mathbf{x}, 0) = \boldsymbol{\omega}_0(\mathbf{x}), \quad \mathbf{m}(\mathbf{x}, 0) = \mathbf{m}_0(\mathbf{x}), \quad \forall \mathbf{x} \in \Omega, \quad (3)$$

and the boundary conditions

$$\mathbf{u} = 0, \quad \boldsymbol{\omega} = 0, \quad \mathbf{m} \cdot \mathbf{n} = 0, \quad \mathbf{H} \times \mathbf{n} = 0 \quad \text{on } \Gamma_T, \quad (4)$$

where $\mathbf{u}_0, \boldsymbol{\omega}_0$, and $\mathbf{m}_0$ are prescribed initial data.

As established in [34], the solution of the Rosensweig system satisfies the following regularity estimates under suitable assumptions on the data and the domain.

Lemma 3.1 (Regularity estimates for the Rosensweig system). *Let $(\mathbf{u}, p, \boldsymbol{\omega}, \mathbf{m}, \mathbf{H})$ be a solution of the Rosensweig ferrofluid flow problem (2). Following the regularity framework established in [34, Theorems 3.2–3.6], assume that the following conditions hold.*

(i) *The external magnetic field satisfies*

$$\mathbf{H}_e \in H^1(0, T; (H^2(\Omega))^3) \cap H^2(0, T; (L^2(\Omega))^3) \cap L^\infty(0, T; (H^2(\Omega))^3). \quad (5)$$

(ii) *The initial data satisfy*

$$\begin{aligned} \mathbf{u}_0 &\in \mathbf{S} \cap H_0(\operatorname{div} 0; \Omega) \cap (H^2(\Omega))^3, \\ \boldsymbol{\omega}_0 &\in \mathbf{S} \cap (H^2(\Omega))^3, \\ \mathbf{m}_0 &\in \mathbf{V} \cap (H^2(\Omega))^3. \end{aligned} \quad (6)$$

(iii) *The solution satisfies the boundedness condition*

$$\int_0^T (\|\mathbf{u}(t)\|_1^4 + \|\mathbf{m}(t)\|_n^4) dt \leq C, \quad (7)$$

where

$$\|\mathbf{v}\|_n^2 := \|\operatorname{curl} \mathbf{v}\|^2 + \|\operatorname{div} \mathbf{v}\|^2.$$

(iv) *Whenever the higher spatial regularity estimates below are invoked, the domain Ω is assumed to possess sufficient elliptic regularity in the following sense.*

- *For any $\mathbf{f} \in (L^2(\Omega))^3$, the unique solution $(\mathbf{v}, q)$ of the Stokes problem*

$$\begin{cases} -\Delta \mathbf{v} + \nabla q = \mathbf{f}, & \text{in } \Omega, \\ \operatorname{div} \mathbf{v} = 0, & \text{in } \Omega, \\ \mathbf{v} = \mathbf{0}, & \text{on } \partial\Omega, \\ \int_\Omega q dx = 0, \end{cases} \quad (8)$$

satisfies

$$\|\mathbf{v}\|_2 + \|q\|_1 \lesssim \|\mathbf{f}\|. \quad (9)$$

- For any $\mathbf{g} \in (L^2(\Omega))^3$, the unique solution $\mathbf{s} \in (H_0^1(\Omega))^3$ of the vector elliptic problem

$$\begin{cases} -\Delta \mathbf{s} - \left(1 + \frac{\lambda'}{\eta'}\right) \nabla \operatorname{div} \mathbf{s} = \mathbf{g}, & \text{in } \Omega, \\ \mathbf{s} = \mathbf{0}, & \text{on } \partial\Omega, \end{cases} \quad (10)$$

satisfies

$$\|\mathbf{s}\|_2 \lesssim \|\mathbf{g}\|. \quad (11)$$

- For any $\mathbf{j} \in (L^2(\Omega))^3$, the unique solution $\mathbf{F}$ of the Maxwell-type problem

$$\begin{cases} \operatorname{curl} \operatorname{curl} \mathbf{F} - \nabla \operatorname{div} \mathbf{F} = \mathbf{j}, & \text{in } \Omega, \\ \operatorname{curl} \mathbf{F} \times \mathbf{n} = \mathbf{0}, & \text{on } \partial\Omega, \\ \mathbf{F} \cdot \mathbf{n} = 0, & \text{on } \partial\Omega, \end{cases} \quad (12)$$

satisfies

$$\|\mathbf{F}\|_2 \lesssim \|\mathbf{j}\|. \quad (13)$$

Then the following estimates hold. First, under assumptions (i)–(iii),

$$\sup_{0 \leq t \leq T} (\|\nabla \mathbf{u}(t)\|^2 + \|\nabla \boldsymbol{\omega}(t)\|^2 + \|\mathbf{m}(t)\|_{\mathbf{n}}^2 + \|\mathbf{H}(t)\|_{\mathbf{n}}^2) + \int_0^T (\|\mathbf{u}\|_2^2 + \|\boldsymbol{\omega}\|_2^2 + \|\mathbf{m}\|_2^2 + \|\mathbf{H}\|_2^2) dt \leq C, \quad (14)$$

and

$$\sup_{0 \leq t \leq T} (\|\partial_t \mathbf{u}(t)\|^2 + \|\partial_t \boldsymbol{\omega}(t)\|^2 + \|\partial_t \mathbf{m}(t)\|_{\mathbf{n}}^2 + \|\partial_t \mathbf{H}(t)\|_{\mathbf{n}}^2) + \int_0^T (\|\nabla \partial_t \mathbf{u}\|^2 + \|\nabla \partial_t \boldsymbol{\omega}\|^2 + \|\partial_t \mathbf{m}\|_{\mathbf{n}}^2 + \|\partial_t \mathbf{H}\|_{\mathbf{n}}^2) dt \leq C. \quad (15)$$

If, in addition, assumption (iv) holds, then

$$\sup_{0 \leq t \leq T} (\|\mathbf{u}(t)\|_2 + \|\boldsymbol{\omega}(t)\|_2 + \|\mathbf{m}(t)\|_2 + \|\mathbf{H}(t)\|_2 + \|p(t)\|_1) \leq C. \quad (16)$$

Moreover, if the initial data satisfy the stronger regularity

$$\begin{aligned} \mathbf{u}_0 &\in \mathbf{S} \cap H_0(\operatorname{div} 0; \Omega) \cap (H^3(\Omega))^3, \\ \boldsymbol{\omega}_0 &\in \mathbf{S} \cap (H^3(\Omega))^3, \\ \mathbf{m}_0 &\in \mathbf{V} \cap (H^3(\Omega))^3, \end{aligned} \quad (17)$$

then

$$\sup_{0 \leq t \leq T} (\|\nabla \partial_t \mathbf{u}(t)\|^2 + \|\nabla \partial_t \boldsymbol{\omega}(t)\|^2 + \|\partial_t \mathbf{m}(t)\|_{\mathbf{n}}^2 + \|\partial_t \mathbf{H}(t)\|_{\mathbf{n}}^2) + \int_0^T (\|\partial_{tt} \mathbf{u}\|^2 + \|\partial_{tt} \boldsymbol{\omega}\|^2 + \|\partial_{tt} \mathbf{m}\|^2 + \|\partial_{tt} \mathbf{H}\|^2) dt \leq C, \quad (18)$$

and

$$\int_0^T (\|\partial_t \mathbf{u}\|_2^2 + \|\partial_t p\|_1^2 + \|\partial_t \boldsymbol{\omega}\|_2^2 + \|\partial_t \mathbf{m}\|_2^2 + \|\partial_t \mathbf{H}\|_2^2) dt \leq C. \quad (19)$$

3.1 Reformulation of the governing equations

Equation (2e), together with the boundary condition (4), implies the existence of a scalar magnetic potential $\varphi \in H_0^1(\Omega)$ such that

$$\mathbf{H} = \nabla \varphi.$$

Consequently, (2f) can be rewritten as

$$\mu_0 \Delta \varphi + \mu_0 \operatorname{div} \mathbf{m} = -\operatorname{div} \mathbf{H}_e. \quad (20)$$

Differentiating (20) with respect to time and using (2d) together with (2f), we obtain

$$\begin{aligned} & -\mu_0 \Delta \varphi_t - \mu_0 \operatorname{div}(\boldsymbol{\omega} \times \mathbf{m}) - \frac{\chi_0}{\tau} \operatorname{div} \mathbf{m} - \frac{\mu_0(1 + \chi_0) + \chi_0}{\tau} \Delta \varphi + \mu_0 \operatorname{div}((\mathbf{u} \cdot \nabla) \mathbf{m}) + \mu_0 \sigma \Delta^2 \varphi \\ & = \operatorname{div} \partial_t \mathbf{H}_e + \frac{\mu_0 + \chi_0}{\tau \mu_0} \operatorname{div} \mathbf{H}_e - \sigma \operatorname{div} \operatorname{grad} \operatorname{div} \mathbf{H}_e. \end{aligned} \quad (21)$$

The initial value $\varphi_0 := \varphi(\cdot, 0) \in H_0^1(\Omega)$ is determined by

$$-\mu_0 \Delta \varphi_0 = \mu_0 \operatorname{div} \mathbf{m}_0 + \operatorname{div} \mathbf{H}_e(\cdot, 0).$$

To facilitate the construction of a structure-preserving weak formulation, we introduce the auxiliary variables

$$\mathbf{z} := \mathbf{u} \times \mathbf{m}, \quad \mathbf{k} := \operatorname{curl} \mathbf{m}, \quad \tilde{p} := p - \frac{\mu_0}{2} \mathbf{m} \cdot \mathbf{H} = p - \frac{\mu_0}{2} \mathbf{m} \cdot \nabla \varphi, \quad \vartheta := -\Delta \varphi, \quad \varpi = \mathbf{u} \cdot \mathbf{m} \quad (22)$$

and use the vector identities

$$\begin{cases} (\mathbf{u} \cdot \nabla) \mathbf{m} = \frac{1}{2} \mathbf{u} \operatorname{div} \mathbf{m} - \frac{1}{2} \mathbf{m} \operatorname{div} \mathbf{u} + \frac{1}{2} \nabla(\mathbf{u} \cdot \mathbf{m}) + \frac{1}{2} \operatorname{curl} \mathbf{u} \times \mathbf{m} + \frac{1}{2} \operatorname{curl} \mathbf{m} \times \mathbf{u} + \frac{1}{2} \operatorname{curl}(\mathbf{m} \times \mathbf{u}), \\ (\mathbf{m} \cdot \nabla) \mathbf{H} = \frac{1}{2} \mathbf{m} \operatorname{div} \mathbf{H} - \frac{1}{2} \mathbf{H} \operatorname{div} \mathbf{m} + \frac{1}{2} \nabla(\mathbf{m} \cdot \mathbf{H}) + \frac{1}{2} \operatorname{curl} \mathbf{m} \times \mathbf{H} + \frac{1}{2} \operatorname{curl} \mathbf{H} \times \mathbf{m} + \frac{1}{2} \operatorname{curl}(\mathbf{H} \times \mathbf{m}), \end{cases}$$

to obtain the following weak formulation of (2a)–(2f): find

$$\begin{aligned} \mathbf{u} : [0, T] &\rightarrow \mathbf{S}, \quad \tilde{p} : [0, T] \rightarrow W, \quad \boldsymbol{\omega} : [0, T] \rightarrow \mathbf{S}, \quad \mathbf{m} : [0, T] \rightarrow \mathbf{V}, \\ \mathbf{z} : [0, T] &\rightarrow \mathbf{U}, \quad \mathbf{k} : [0, T] \rightarrow \mathbf{U}, \quad \varphi : [0, T] \rightarrow \Sigma, \quad \vartheta : [0, T] \rightarrow \Sigma, \quad \varpi : [0, T] \rightarrow \Sigma, \end{aligned}$$

such that, for all

$$\mathbf{v} \in \mathbf{S}, \quad q \in W, \quad \mathbf{s} \in \mathbf{S}, \quad \mathbf{F} \in \mathbf{V}, \quad \boldsymbol{\lambda} \in \mathbf{U}, \quad \boldsymbol{\theta} \in \mathbf{U}, \quad \psi \in \Sigma, \quad \varsigma \in \Sigma, \quad \gamma \in \Sigma,$$

the following equations hold:

$$\left\{ \begin{aligned} &\rho(\partial_t \mathbf{u}, \mathbf{v}) + \rho b(\mathbf{u}, \mathbf{u}, \mathbf{v}) + \eta(\nabla \mathbf{u}, \nabla \mathbf{v}) + \zeta(\operatorname{curl} \mathbf{u}, \operatorname{curl} \mathbf{v}) - 2\zeta(\boldsymbol{\omega}, \operatorname{curl} \mathbf{v}) - (\tilde{p}, \operatorname{div} \mathbf{v}) + \frac{\mu_0}{2}(\nabla \varphi \operatorname{div} \mathbf{m}, \mathbf{v}) \\ &\quad + \frac{\mu_0}{2}(\vartheta, \mathbf{m} \cdot \mathbf{v}) - \frac{\mu_0}{2}(\mathbf{v} \times \mathbf{k}, \nabla \varphi) - \frac{\mu_0}{2}(\mathbf{m} \times \operatorname{curl} \mathbf{v}, \nabla \varphi) = 0, \\ &\quad (\operatorname{div} \mathbf{u}, q) = 0, \\ &\rho \kappa(\partial_t \boldsymbol{\omega}, \mathbf{s}) + \eta'(\nabla \boldsymbol{\omega}, \nabla \mathbf{s}) + (\eta' + \lambda')(\operatorname{div} \boldsymbol{\omega}, \operatorname{div} \mathbf{s}) + 4\zeta(\boldsymbol{\omega}, \mathbf{s}) + \rho \kappa b(\mathbf{u}, \boldsymbol{\omega}, \mathbf{s}) - \mu_0(\mathbf{m} \times \nabla \varphi, \mathbf{s}) - 2\zeta(\operatorname{curl} \mathbf{u}, \mathbf{s}) \\ &\quad (\partial_t \mathbf{m}, \mathbf{F}) + \frac{1}{\tau}(\mathbf{m}, \mathbf{F}) + \sigma(\operatorname{curl} \mathbf{k}, \mathbf{F}) + \sigma(\operatorname{div} \mathbf{m}, \operatorname{div} \mathbf{F}) - c(\mathbf{u}, \mathbf{m}, \mathbf{F}) - \frac{1}{2}(\mathbf{m} \times \operatorname{curl} \mathbf{u}, \mathbf{F}) - \frac{1}{2}(\mathbf{u} \times \mathbf{k}, \mathbf{F}) \\ &\quad - \frac{1}{2}(\operatorname{curl} \mathbf{z}, \mathbf{F}) - (\boldsymbol{\omega} \times \mathbf{m}, \mathbf{F}) - \frac{\chi_0}{\tau}(\nabla \varphi, \mathbf{F}) = 0, \\ &\quad (\mathbf{z}, \boldsymbol{\lambda}) - (\mathbf{u} \times \mathbf{m}, \boldsymbol{\lambda}) = 0, \\ &\quad (\mathbf{k}, \boldsymbol{\theta}) - (\mathbf{m}, \operatorname{curl} \boldsymbol{\theta}) = 0, \\ &\mu_0(\partial_t \nabla \varphi, \nabla \psi) + \frac{\mu_0(1+\chi_0)+\chi_0}{\tau}(\nabla \varphi, \nabla \psi) + \sigma \mu_0(\nabla \vartheta, \nabla \psi) + \mu_0(\boldsymbol{\omega} \times \mathbf{m}, \nabla \psi) + \frac{\chi_0}{\tau}(\mathbf{m}, \nabla \psi) - \frac{\mu_0}{2}(\mathbf{u} \operatorname{div} \mathbf{m}, \nabla \psi) \\ &\quad - \frac{\mu_0}{2}(\nabla \varpi, \nabla \psi) + \frac{\mu_0}{2}(\mathbf{m} \times \operatorname{curl} \mathbf{u}, \nabla \psi) + \frac{\mu_0}{2}(\mathbf{u} \times \mathbf{k}, \nabla \psi) \\ &\quad + (\partial_t \mathbf{H}_e, \nabla \psi) + \frac{\mu_0+\chi_0}{\tau \mu_0}(\mathbf{H}_e, \nabla \psi) - \sigma(\operatorname{grad} \operatorname{div} \mathbf{H}_e, \nabla \psi) = 0, \\ &\quad \sigma \mu_0(\vartheta, \varsigma) - \sigma \mu_0(\nabla \varphi, \nabla \varsigma) = 0, \\ &\quad (\varpi, \gamma) - (\mathbf{u} \cdot \mathbf{m}, \gamma) = 0, \end{aligned} \right. \quad (23)$$

The system is supplemented with the initial data (3), where

$$\begin{aligned} b(\mathbf{u}, \mathbf{v}, \mathbf{w}) &:= \frac{1}{2} [((\mathbf{u} \cdot \nabla) \mathbf{v}, \mathbf{w}) - ((\mathbf{u} \cdot \nabla) \mathbf{w}, \mathbf{v})], \\ c(\mathbf{v}, \mathbf{m}, \mathbf{F}) &:= \frac{1}{2} [(\mathbf{m} \cdot \mathbf{v}, \operatorname{div} \mathbf{F}) - (\mathbf{F} \cdot \mathbf{v}, \operatorname{div} \mathbf{m})]. \end{aligned}$$

3.2 An abstract formulation

To reveal the underlying skew-symmetric structure of the coupled system and to facilitate the construction and analysis of energy-stable time discretizations, we next recast the weak formulation (23) in an abstract form.

For

$$\begin{aligned} \mathcal{X} &= (\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \varphi)^T \in \mathbb{V}, \quad \mathcal{Y} = (\mathbf{v}, \mathbf{s}, \mathbf{F}, \psi)^T \in \mathbb{V}, \\ \mathcal{Z} &= (\boldsymbol{\epsilon}, \boldsymbol{\sigma}, \mathbf{A}, \phi)^T \in \mathbb{V}, \quad q \in W, \end{aligned}$$

we define the bilinear forms

$$\mathbf{D} : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{R}, \quad \mathbf{A} : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{R}, \quad \mathbf{C} : \mathbb{V} \times W \rightarrow \mathbb{R}$$

by

$$\begin{aligned} \mathbf{D}(\mathcal{X}, \mathcal{Y}) &:= \rho(\mathbf{u}, \mathbf{v}) + \rho \kappa(\boldsymbol{\omega}, \mathbf{s}) + (\mathbf{m}, \mathbf{F}) + \mu_0(\nabla \varphi, \nabla \psi), \\ \mathbf{A}(\mathcal{X}, \mathcal{Y}) &:= \eta(\nabla \mathbf{u}, \nabla \mathbf{v}) + \eta'(\nabla \boldsymbol{\omega}, \nabla \mathbf{s}) + (\eta' + \lambda')(\operatorname{div} \boldsymbol{\omega}, \operatorname{div} \mathbf{s}) + \sigma(\operatorname{curl} \mathbf{m}, \operatorname{curl} \mathbf{F}) + \sigma(\operatorname{div} \mathbf{m}, \operatorname{div} \mathbf{F}) + \frac{1}{\tau}(\mathbf{m}, \mathbf{F}) \\ &\quad + \frac{\mu_0(1+\chi_0)+\chi_0}{\tau}(\nabla \varphi, \nabla \psi) + \sigma \mu_0(\Delta \varphi, \Delta \psi) + \zeta(\operatorname{curl} \mathbf{u}, \operatorname{curl} \mathbf{v}) - 2\zeta(\boldsymbol{\omega}, \operatorname{curl} \mathbf{v}) - 2\zeta(\operatorname{curl} \mathbf{u}, \mathbf{s}) + 4\zeta(\boldsymbol{\omega}, \mathbf{s}), \end{aligned}$$

and

$$\mathbf{C}(\mathcal{Y}, q) := -(\operatorname{div} \mathbf{v}, q).$$

We further define the trilinear form

$$\mathbf{B} : \mathbb{V} \times \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{R}$$

by

$$\begin{aligned} \mathbf{B}(\mathcal{Z}, \mathcal{X}, \mathcal{Y}) := & \rho b(\boldsymbol{\epsilon}, \mathbf{u}, \mathbf{v}) + \rho \kappa b(\boldsymbol{\epsilon}, \boldsymbol{\omega}, \mathbf{s}) - c(\boldsymbol{\epsilon}, \mathbf{m}, \mathbf{F}) - \frac{1}{2}(\mathbf{m} \times \operatorname{curl} \boldsymbol{\epsilon}, \mathbf{F}) - (\boldsymbol{\sigma} \times \mathbf{m}, \mathbf{F}) + \mu_0 [(\nabla \varphi, \mathbf{v} \operatorname{div} \mathbf{A}) - (\nabla \psi, \mathbf{u} \operatorname{div} \mathbf{A})] \\ & + \frac{\mu_0}{2} [(\mathbf{u} \times \operatorname{curl} \mathbf{A}, \nabla \psi) - (\mathbf{v} \times \operatorname{curl} \mathbf{A}, \nabla \varphi)] + \frac{\mu_0}{2} [(\mathbf{A} \times \operatorname{curl} \mathbf{u}, \nabla \psi) - (\mathbf{A} \times \operatorname{curl} \mathbf{v}, \nabla \varphi)] \\ & + \mu_0 [(\boldsymbol{\omega} \times \mathbf{A}, \nabla \psi) - (\mathbf{s} \times \mathbf{A}, \nabla \varphi)] + \frac{\chi_0}{\tau} [(\mathbf{m}, \nabla \psi) - (\mathbf{F}, \nabla \varphi)] + \frac{1}{2} [(\boldsymbol{\epsilon} \times \mathbf{F}, \operatorname{curl} \mathbf{m}) - (\boldsymbol{\epsilon} \times \mathbf{m}, \operatorname{curl} \mathbf{F})] \\ & + \frac{\mu_0}{2} [(Q^\Sigma \Delta \psi, \mathbf{u} \cdot \mathbf{A}) - (Q^\Sigma \Delta \varphi, \mathbf{v} \cdot \mathbf{A})], \end{aligned}$$

where $Q^\Sigma : L^2(\Omega) \rightarrow \Sigma$ is the L^2 orthogonal projection operator.

The definitions of the above forms immediately yield the following structural properties.

Lemma 3.2. *The bilinear forms $\mathbf{A}$ and $\mathbf{C}$ and the trilinear form $\mathbf{B}$ satisfy the following properties:*

(1) *The bilinear form $\mathbf{A}$ is bounded, symmetric, and coercive; namely,*

$$|\mathbf{A}(\mathcal{X}, \mathcal{Y})| \lesssim \|\mathcal{X}\|_{\mathbb{V}} \|\mathcal{Y}\|_{\mathbb{V}}, \quad \forall \mathcal{X}, \mathcal{Y} \in \mathbb{V}, \quad (24)$$

$$\mathbf{A}(\mathcal{X}, \mathcal{Y}) = \mathbf{A}(\mathcal{Y}, \mathcal{X}), \quad \forall \mathcal{X}, \mathcal{Y} \in \mathbb{V}, \quad (25)$$

$$\mathbf{A}(\mathcal{Y}, \mathcal{Y}) \gtrsim \|\mathcal{Y}\|_{\mathbb{V}}^2, \quad \forall \mathcal{Y} \in \mathbb{V}. \quad (26)$$

(2) *The bilinear form $\mathbf{C}$ is bounded:*

$$|\mathbf{C}(\mathcal{Y}, q)| \lesssim \|\mathcal{Y}\|_{\mathbb{V}} \|q\|, \quad \forall \mathcal{Y} \in \mathbb{V}, \quad q \in W, \quad (27)$$

and satisfies the inf-sup condition

$$\sup_{\mathcal{Y} \in \mathbb{V} \setminus \{0\}} \frac{|\mathbf{C}(\mathcal{Y}, q)|}{\|\mathcal{Y}\|_{\mathbb{V}}} \gtrsim \|q\|, \quad \forall q \in W. \quad (28)$$

(3) *The trilinear form $\mathbf{B}$ is skew-symmetric with respect to its second and third arguments:*

$$\mathbf{B}(\mathcal{Z}, \mathcal{X}, \mathcal{Y}) = -\mathbf{B}(\mathcal{Z}, \mathcal{Y}, \mathcal{X}), \quad \forall \mathcal{X}, \mathcal{Y}, \mathcal{Z} \in \mathbb{V}. \quad (29)$$

In particular,

$$\mathbf{B}(\mathcal{Z}, \mathcal{Y}, \mathcal{Y}) = 0, \quad \forall \mathcal{Y}, \mathcal{Z} \in \mathbb{V}.$$

Proof. The estimates (24), (25), (26), (27), and (29) follow directly from the definitions of the corresponding forms. It therefore remains only to verify (28).

For any $q \in W = L_0^2(\Omega)$, the standard inf-sup result implies that there exists $\mathbf{v} \in \mathbf{S} = (H_0^1(\Omega))^3$ such that

$$\operatorname{div} \mathbf{v} = q, \quad \|\nabla \mathbf{v}\| \lesssim \|q\|.$$

Taking

$$\mathcal{Z} = (-\mathbf{v}, \mathbf{0}, \mathbf{0}, 0)^\top \in \mathbb{V},$$

we obtain

$$\sup_{\mathcal{Y} \in \mathbb{V} \setminus \{0\}} \frac{|\mathbf{C}(\mathcal{Y}, q)|}{\|\mathcal{Y}\|_{\mathbb{V}}} \geq \frac{|\mathbf{C}(\mathcal{Z}, q)|}{\|\mathcal{Z}\|_{\mathbb{V}}} = \frac{(\operatorname{div} \mathbf{v}, q)}{\|\nabla \mathbf{v}\|} \gtrsim \|q\|,$$

which proves (28). □

Remark 1. *The bilinear form $\mathbf{A}$ induces the norm*

$$\|\mathcal{Y}\|_{\mathbf{A}} := \mathbf{A}(\mathcal{Y}, \mathcal{Y})^{1/2}, \quad \forall \mathcal{Y} \in \mathbb{V}.$$

By Lemma 3.2, this norm is equivalent to the norm of $\mathbb{V}$:

$$\|\mathcal{Y}\|_{\mathbb{V}} \simeq \|\mathcal{Y}\|_{\mathbf{A}}, \quad \forall \mathcal{Y} \in \mathbb{V}.$$

With the above notation, the weak formulation (23) can be expressed compactly in the following abstract form: find

$$\mathcal{X} = (\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \varphi)^\top : [0, T] \rightarrow \mathbb{V}$$

and $\tilde{p} : [0, T] \rightarrow W$ such that

$$\begin{cases} \mathbb{D}(\partial_t \mathcal{X}, \mathcal{Y}) + \mathbf{A}(\mathcal{X}, \mathcal{Y}) + \mathbf{B}(\mathcal{X}, \mathcal{X}, \mathcal{Y}) + \mathbf{C}(\mathcal{Y}, \tilde{p}) &= \langle \mathbf{f}, \mathcal{Y} \rangle, & \forall \mathcal{Y} \in \mathbb{V}, \\ \mathbf{C}(\mathcal{X}, q) &= 0, & \forall q \in W, \end{cases} \quad (30)$$

with the initial condition

$$\mathcal{X}(0) = \mathcal{X}_0 := (\mathbf{u}_0, \boldsymbol{\omega}_0, \mathbf{m}_0, \varphi_0)^\top,$$

where

$$\mathbf{f} = \left(\mathbf{0}, \mathbf{0}, \mathbf{0}, \operatorname{div} \partial_t \mathbf{H}_e + \frac{\mu_0 + \chi_0}{\tau \mu_0} \operatorname{div} \mathbf{H}_e - \sigma \operatorname{div} \operatorname{grad} \operatorname{div} \mathbf{H}_e \right)^\top.$$

The following result establishes the equivalence between the original weak formulation and the abstract formulation.

Theorem 3.3. *The weak formulation (23) and the abstract formulation (30) are equivalent in the following sense:*

(i) If

$$\mathbf{u} \in \mathbf{S}, \quad \tilde{p} \in W, \quad \boldsymbol{\omega} \in \mathbf{S}, \quad \mathbf{m} \in \mathbf{V} \cap H(\operatorname{curl}; \Omega),$$

and

$$\mathbf{z}, \mathbf{k} \in \mathbf{U}, \quad \varphi \in D(-\Delta), \quad \vartheta \in D(-\Delta), \quad \varpi \in \Sigma,$$

satisfy (23), then

$$\mathcal{X} = (\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \varphi)^\top$$

and $\tilde{p}$ satisfy (30).

(ii) Conversely, if

$$\mathcal{X} = (\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \varphi)^\top$$

and $\tilde{p}$ satisfy (30), then, with

$$\mathbf{z} = Q^c(\mathbf{u} \times \mathbf{m}), \quad \mathbf{k} = \operatorname{curl} \mathbf{m}, \quad \varpi = Q^\Sigma(\mathbf{u} \cdot \mathbf{m}),$$

the variables $\mathbf{u}, \tilde{p}, \boldsymbol{\omega}, \mathbf{m}, \mathbf{z}, \mathbf{k}, \varphi, \vartheta$, and ϖ satisfy the weak formulation (23).

3.3 Energy estimates

We conclude this section by deriving an energy estimate for the weak formulation (23), or equivalently, for the abstract system (30). The skew-symmetry of the trilinear form $\mathbf{B}$ plays a fundamental role in the analysis. We shall also use the following elementary inequality.

Lemma 3.4. [17, Lemma 1] *Let $x \geq 0$ satisfy*

$$x^2 \leq \gamma^2 + \beta x$$

for some $\beta, \gamma \geq 0$. Then

$$x \leq \beta + \gamma.$$

For any $\mathcal{Y} = (\mathbf{v}, \mathbf{s}, \mathbf{F}, \psi)^\top \in \mathbb{V}$ and $\mathbf{f} \in \mathbb{V}'$, define

$$\|\mathcal{Y}\|_{\mathbb{D}} := \mathbb{D}(\mathcal{Y}, \mathcal{Y})^{1/2}$$

and the associated dual norm

$$\|\mathbf{f}\|_{\mathbb{D}^{-1}} := \sup_{\mathcal{Y} \in \mathbb{V} \setminus \{0\}} \frac{\langle \mathbf{f}, \mathcal{Y} \rangle}{\|\mathcal{Y}\|_{\mathbb{D}}}.$$

The following theorem establishes the energy stability of the continuous system.

Theorem 3.5. *Assume that*

$$\mathbf{H}_e \in W^{1,1}(L^2(\Omega)) \cap L^1(H(\operatorname{div})),$$

and let $\mathcal{X}$ be a weak solution of (30). Then

$$\left(\sup_{0 \leq t \leq T} \|\mathcal{X}(\cdot, t)\|_{\mathbb{D}}^2 + 2 \int_0^T \|\mathcal{X}(\cdot, s)\|_{\mathbf{A}}^2 \, ds \right)^{1/2} \leq \|\mathcal{X}_0\|_{\mathbb{D}} + 2 \int_0^T \|\mathbf{f}(\cdot, s)\|_{\mathbb{D}^{-1}} \, ds. \quad (31)$$

Proof. Taking $\mathcal{Y} = \mathcal{X}$ in (30), using the constraint $\mathcal{C}(\mathcal{X}, \bar{p}) = 0$, and exploiting the skew-symmetry

$$\mathcal{B}(\mathcal{X}, \mathcal{X}, \mathcal{X}) = 0,$$

we obtain

$$\frac{1}{2} \frac{d}{dt} \|\mathcal{X}\|_{\mathbb{D}}^2 + \|\mathcal{X}\|_{\mathbb{A}}^2 = \langle \mathbf{f}, \mathcal{X} \rangle \leq \|\mathbf{f}\|_{\mathbb{D}^{-1}} \|\mathcal{X}\|_{\mathbb{D}}.$$

Integrating over $(0, t)$, for any $t \in (0, T]$, yields

$$\begin{aligned} \|\mathcal{X}(\cdot, t)\|_{\mathbb{D}}^2 + 2 \int_0^t \|\mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds &\leq \|\mathcal{X}_0\|_{\mathbb{D}}^2 + 2 \int_0^t \|\mathbf{f}(\cdot, s)\|_{\mathbb{D}^{-1}} \|\mathcal{X}(\cdot, s)\|_{\mathbb{D}} ds \\ &\leq \|\mathcal{X}_0\|_{\mathbb{D}}^2 + 2 \sup_{0 \leq s \leq t} \|\mathcal{X}(\cdot, s)\|_{\mathbb{D}} \int_0^t \|\mathbf{f}(\cdot, s)\|_{\mathbb{D}^{-1}} ds. \end{aligned}$$

Taking the supremum over $t \in [0, T]$ and applying Lemma 3.4 gives (31). $\square$

The abstract formulation developed above isolates the symmetric dissipative part of the system from its skew-symmetric nonlinear couplings, thereby providing a convenient framework for the construction of energy-stable time discretizations. In the remainder of this paper, we exploit this structure to develop and analyze first- and second-order semi-implicit schemes for the Rosensweig model. The temporal discretizations are designed to retain the continuous energy-dissipation mechanism while treating the nonlinear coupling terms explicitly or semi-explicitly, so that only linear systems need to be solved at most time levels. For the spatial discretization, compatible H^1 -, $H(\text{curl})$ -, and $H(\text{div})$ -conforming finite element spaces are employed within the framework of finite element exterior calculus, thereby preserving the underlying differential structure at the discrete level.

4 First-Order Scheme

In this section, we develop and analyze a first-order semi-implicit finite element scheme for the ferrohydrodynamic equations. We first introduce a time semi-discrete scheme based on a modified implicit Euler discretization and establish its well-posedness, energy stability, and temporal error estimates. We then construct a compatible spatial discretization within the framework of finite element exterior calculus, derive the corresponding fully discrete scheme, and present an exact Schur complement decoupling strategy for its efficient solution. Finally, we introduce the consistency residuals required for the fully discrete error analysis.

4.1 Time semi-discretization

For each $1 \leq n \leq N$, we consider the following semi-implicit time discretization of (30).

Algorithm 1 First-order scheme (time semi-discretization)

Given $\mathcal{X}^{n-1} \in \mathbb{V}$, find $\mathcal{X}^n \in \mathbb{V}$ and $\tilde{p}^n \in W$ such that

$$\begin{cases} \mathcal{D}(\delta_t \mathcal{X}^n, \mathcal{Y}) + \mathcal{A}(\mathcal{X}^n, \mathcal{Y}) + \mathcal{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathcal{Y}) + \mathcal{C}(\mathcal{Y}, \tilde{p}^n) = \langle \mathbf{f}(\cdot, t_n), \mathcal{Y} \rangle, & \forall \mathcal{Y} \in \mathbb{V}, \\ \mathcal{C}(\mathcal{X}^n, q) = 0, & \forall q \in W, \end{cases} \quad (32)$$

with initial condition

$$\mathcal{X}^0 = \mathcal{X}_0. \quad (33)$$

The following theorem establishes the well-posedness of the time semi-discrete scheme Algorithm 1.

Theorem 4.1. *For any $1 \leq n \leq N$, given $\mathcal{X}^{n-1} \in \mathbb{V}$, the time semi-discrete scheme Algorithm 1 admits a unique solution $(\mathcal{X}^n, \tilde{p}^n) \in \mathbb{V} \times W$.*

Proof. The scheme Algorithm 1 can equivalently be written as

$$\begin{cases} \mathcal{D}(\mathcal{X}^n, \mathcal{Y}) + \Delta t \mathcal{A}(\mathcal{X}^n, \mathcal{Y}) + \Delta t \mathcal{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathcal{Y}) + \Delta t \mathcal{C}(\mathcal{Y}, \tilde{p}^n) = \mathcal{D}(\mathcal{X}^{n-1}, \mathcal{Y}) + \Delta t \langle \mathbf{f}(\cdot, t_n), \mathcal{Y} \rangle, & \forall \mathcal{Y} \in \mathbb{V}, \\ \Delta t \mathcal{C}(\mathcal{X}^n, q) = 0, & \forall q \in W. \end{cases}$$

By lemma 3.2, the left-hand-side bilinear form satisfies the boundedness estimate

$$|\mathcal{D}(\mathcal{X}^n, \mathcal{Y}) + \Delta t \mathbf{A}(\mathcal{X}^n, \mathcal{Y}) + \Delta t \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathcal{Y})| \lesssim (\|\mathcal{X}^n\|_{\mathbb{D}}^2 + \Delta t \|\mathcal{X}^n\|_{\mathbb{V}}^2)^{1/2} (\|\mathcal{Y}\|_{\mathbb{D}}^2 + \Delta t \|\mathcal{Y}\|_{\mathbb{V}}^2)^{1/2},$$

and the coercivity estimate

$$\mathcal{D}(\mathcal{Y}, \mathcal{Y}) + \Delta t \mathbf{A}(\mathcal{Y}, \mathcal{Y}) + \Delta t \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{Y}, \mathcal{Y}) \geq \|\mathcal{Y}\|_{\mathbb{D}}^2 + \Delta t \|\mathcal{Y}\|_{\mathbb{V}}^2, \quad \forall \mathcal{Y} \in \mathbb{V}.$$

Define the subspace

$$\mathfrak{K} = \{\mathcal{Y} \in \mathbb{V} : \mathbf{C}(\mathcal{Y}, q) = 0, \quad \forall q \in W\}.$$

Restricting the problem to $\mathfrak{K}$, the Lax–Milgram theorem [9] guarantees the existence of a unique $\mathcal{X}^n \in \mathfrak{K}$ satisfying

$$\mathcal{D}(\mathcal{X}^n, \mathcal{Y}) + \Delta t \mathbf{A}(\mathcal{X}^n, \mathcal{Y}) + \Delta t \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathcal{Y}) = \mathcal{D}(\mathcal{X}^{n-1}, \mathcal{Y}) + \Delta t (\mathbf{f}(\cdot, t_n), \mathcal{Y}), \quad \forall \mathcal{Y} \in \mathfrak{K}.$$

Finally, the inf-sup condition (28) ensures the existence of a unique $\tilde{p}^n \in W$ such that

$$\Delta t \mathbf{C}(\mathcal{Y}, \tilde{p}^n) = \mathcal{D}(\mathcal{X}^{n-1}, \mathcal{Y}) + \Delta t (\mathbf{f}(\cdot, t_n), \mathcal{Y}) - \mathcal{D}(\mathcal{X}^n, \mathcal{Y}) - \Delta t \mathbf{A}(\mathcal{X}^n, \mathcal{Y}) - \Delta t \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathcal{Y}), \quad \forall \mathcal{Y} \in \mathbb{V}.$$

This proves the existence and uniqueness of $(\mathcal{X}^n, \tilde{p}^n)$ in Algorithm 1. $\square$

To study the stability of the scheme, we next derive a discrete energy estimate. By choosing appropriate test functions and exploiting the skew-symmetric structure of the nonlinear term, we obtain an energy bound that reflects the dissipative structure of the continuous problem.

Theorem 4.2 (Energy estimate for Algorithm 1). *Given $\mathcal{X}^{n-1} \in \mathbb{V}$ for $1 \leq n \leq N$, let $(\mathcal{X}^n, \tilde{p}^n) \in \mathbb{V} \times W$ be the unique solution of the semi-discrete scheme Algorithm 1. Then the following energy estimate holds:*

$$\left(\max_{1 \leq n \leq N} \|\mathcal{X}^n\|_{\mathbb{D}}^2 + \Delta t \sum_{n=1}^N \|\mathcal{X}^n\|_{\mathbb{A}}^2 \right)^{1/2} \leq \|\mathcal{X}^0\|_{\mathbb{D}} + 2\Delta t \sum_{n=1}^N \|\mathbf{f}(\cdot, t_n)\|_{\mathbb{D}^{-1}}. \quad (34)$$

Proof. The second equation of (32) implies that $\mathcal{X}^n \in \mathfrak{K}$. Taking $\mathcal{Y} = \mathcal{X}^n \in \mathfrak{K} \subset \mathbb{V}$ as a test function in the first equation of (32), we obtain

$$\|\mathcal{X}^n\|_{\mathbb{D}}^2 + \Delta t \|\mathcal{X}^n\|_{\mathbb{A}}^2 = \mathcal{D}(\mathcal{X}^{n-1}, \mathcal{X}^n) + \Delta t (\mathbf{f}(\cdot, t_n), \mathcal{X}^n).$$

Applying the Cauchy–Schwarz and Young inequalities yields

$$\mathcal{D}(\mathcal{X}^{n-1}, \mathcal{X}^n) \leq \frac{1}{2} \|\mathcal{X}^n\|_{\mathbb{D}}^2 + \frac{1}{2} \|\mathcal{X}^{n-1}\|_{\mathbb{D}}^2, \quad \Delta t (\mathbf{f}(\cdot, t_n), \mathcal{X}^n) \leq \Delta t \|\mathbf{f}(\cdot, t_n)\|_{\mathbb{D}^{-1}} \|\mathcal{X}^n\|_{\mathbb{D}}.$$

Hence,

$$\|\mathcal{X}^n\|_{\mathbb{D}}^2 - \|\mathcal{X}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\mathcal{X}^n\|_{\mathbb{A}}^2 \leq 2\Delta t \|\mathbf{f}(\cdot, t_n)\|_{\mathbb{D}^{-1}} \|\mathcal{X}^n\|_{\mathbb{D}}.$$

Summing the above inequality from $n = 1$ to L for any $1 \leq L \leq N$, we obtain

$$\|\mathcal{X}^L\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^L \|\mathcal{X}^n\|_{\mathbb{A}}^2 \leq \|\mathcal{X}^0\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^L \|\mathbf{f}(\cdot, t_n)\|_{\mathbb{D}^{-1}} \|\mathcal{X}^n\|_{\mathbb{D}}.$$

Taking the maximum over $1 \leq n \leq N$ and using the arbitrariness of L , we deduce

$$\max_{1 \leq n \leq N} \|\mathcal{X}^n\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^N \|\mathcal{X}^n\|_{\mathbb{A}}^2 \leq \|\mathcal{X}^0\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^N \|\mathbf{f}(\cdot, t_n)\|_{\mathbb{D}^{-1}} \max_{1 \leq n \leq N} \|\mathcal{X}^n\|_{\mathbb{D}}.$$

The desired estimate follows from Lemma 3.4. $\square$

To derive a stronger regularity estimate, we test the first equation of (32) with $\mathcal{Y} = \mathcal{V} \mathcal{J} \mathcal{A}_1 \mathcal{X}^n \in \mathbb{V}$.

Theorem 4.3. *Given $\mathcal{X}^{n-1} \in \mathbb{V}$ for $1 \leq n \leq N$, let $(\mathcal{X}^n, \tilde{p}^n) \in \mathbb{V} \times W$ be the unique solution of the semi-discrete scheme Algorithm 1. Then the following estimates hold:*

Proof. The second equation of (32) implies that $\{\mathcal{X}^n\}_{n=1}^N \subset \mathfrak{K} \subset \mathbb{V}$. We then take $\mathcal{Y} = \mathcal{V} \mathcal{J} \mathcal{A}_1 \mathcal{X}^n \in \mathbb{V}$ as the test function. $\square$

A further energy estimate can be obtained by taking $\mathcal{Y} = \delta_t \mathcal{X}^n$ as the test function.

Theorem 4.4. *Given $\mathcal{X}^{n-1} \in \mathbb{V}$ for $1 \leq n \leq N$, let $(\mathcal{X}^n, \tilde{p}^n) \in \mathbb{V} \times W$ be the unique solution of the semi-discrete scheme Algorithm 1. Then the following energy estimate holds:*

Proof. The second equation of (32) implies that $\{\mathcal{X}^n\}_{n=1}^N \subset \mathfrak{K} \subset \mathbb{V}$. Taking $\mathcal{Y} = \delta_t \mathcal{X}^n \in \mathfrak{K}$ as the test function in the first equation of (32), we obtain

$$\Delta t \|\delta_t \mathcal{X}^n\|_{\mathbb{D}}^2 + \|\mathcal{X}^n\|_{\mathbb{A}}^2 = \Delta t (\mathbf{f}(\cdot, t_n), \delta_t \mathcal{X}^n) + \mathbf{A}(\mathcal{X}^n, \mathcal{X}^{n-1}) - \Delta t \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \delta_t \mathcal{X}^n).$$

Applying Cauchy-Schwarz and Young's inequality yields

$$\|\mathcal{X}^n\|_{\mathbb{A}}^2 - \|\mathcal{X}^{n-1}\|_{\mathbb{A}}^2 + 2\Delta t \|\delta_t \mathcal{X}^n\|_{\mathbb{D}}^2 \leq 2\Delta t (\mathbf{f}(\cdot, t_n), \delta_t \mathcal{X}^n) - 2\Delta t \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \delta_t \mathcal{X}^n)$$

□

With the stability estimate established, we now turn to the temporal error analysis. The error bounds are obtained by combining the discrete stability with consistency estimates. To this end, we first rewrite the continuous weak formulation (30) at the time levels t_n , $1 \leq n \leq N$, in a form that is directly comparable with Algorithm 1.

$$\begin{cases} \mathbf{D}(\delta_t \mathcal{X}(\cdot, t_n), \mathcal{Y}) + \mathbf{A}(\mathcal{X}(\cdot, t_n), \mathcal{Y}) + \mathbf{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \mathcal{Y}) + \mathbf{C}(\mathcal{Y}, \tilde{p}(\cdot, t_n)) \\ \quad = (\mathbf{f}(\cdot, t_n), \mathcal{Y}) + \mathbf{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \mathcal{Y}) & \forall \mathcal{Y} \in \mathbb{V}, \\ \quad \mathbf{C}(\mathcal{X}(\cdot, t_n), q) = 0 & \forall q \in W. \end{cases} \quad (35)$$

Define the errors $\mathbf{e}^n := \mathcal{X}(\cdot, t_n) - \mathcal{X}^n$ and $e_p^n := \tilde{p}(\cdot, t_n) - \tilde{p}^n$. We can now establish the corresponding temporal error estimates.

Theorem 4.5 (Error estimates for Algorithm 1). *Let $(\mathcal{X}, \tilde{p}) : (0, T) \rightarrow \mathbb{V} \times W$ be the solution of the continuous weak formulation (30) and $\{(\mathcal{X}^n, \tilde{p}^n)\}_{n=1}^N \subset \mathbb{V} \times W$ be the unique solution of the semi-discrete scheme Algorithm 1. Then the following error estimates hold:*

$$\max_{1 \leq n \leq N} \|\mathcal{X}(\cdot, t_n) - \mathcal{X}^n\|_{\mathbb{D}} + \Delta t \sum_{n=1}^N \|\mathcal{X}(\cdot, t_n) - \mathcal{X}^n\|_{\mathbb{V}}^2 \leq \Delta t^2 \int_0^T (\|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{V}}^2) ds, \quad (36)$$

$$\max_{1 \leq n \leq N} \|\mathcal{X}(\cdot, t_n) - \mathcal{X}^n\|_{\mathbb{V}}^2 + \Delta t \sum_{n=1}^N \|\delta_t(\mathcal{X}(\cdot, t_n) - \mathcal{X}^n)\|_{\mathbb{D}}^2 \lesssim \Delta t^2 \int_0^T (\|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2) ds, \quad (37)$$

$$\Delta t \sum_{n=1}^N \|\tilde{p}(\cdot, t_n) - \tilde{p}^n\|^2 \lesssim \Delta t^2 \int_0^T (\|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2) ds. \quad (38)$$

Proof. Subtracting the semi-discrete scheme Algorithm 1 from (35) yields

$$\begin{cases} \mathbf{D}(\delta_t \mathbf{e}^n, \mathcal{Y}) + \mathbf{A}(\mathbf{e}^n, \mathcal{Y}) + \mathbf{C}(\mathcal{Y}, e_p^n) = \mathbf{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \mathcal{Y}) - (\mathbf{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \mathcal{Y}) - \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathcal{Y})) & \forall \mathcal{Y} \in \mathbb{V}, \\ \quad \mathbf{C}(\mathbf{e}^n, q) = 0 & \forall q \in W. \end{cases} \quad (39)$$

Taking $\mathcal{Y} = \mathbf{e}^n \in \mathbb{V}$ and $q = e_p^n \in W$ in the above error equations gives

$$\|\mathbf{e}^n\|_{\mathbb{D}}^2 + \Delta t \|\mathbf{e}^n\|_{\mathbb{A}}^2 = \mathbf{D}(\mathbf{e}^{n-1}, \mathbf{e}^n) + \Delta t \mathbf{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \mathbf{e}^n) - \Delta t (\mathbf{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \mathbf{e}^n) - \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathbf{e}^n)).$$

Applying the Cauchy-Schwarz and Young inequalities yields

$$\|\mathbf{e}^n\|_{\mathbb{D}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\mathbf{e}^n\|_{\mathbb{A}}^2 \leq 2\Delta t |\mathbf{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \mathbf{e}^n)| + 2\Delta t |\mathbf{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \mathbf{e}^n) - \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathbf{e}^n)|.$$

Using the Taylor expansion

$$\mathcal{X}(\cdot, t_{n-1}) = \mathcal{X}(\cdot, t_n) - \Delta t \partial_t \mathcal{X}(\cdot, t_n) + \int_{t_n}^{t_{n-1}} \partial_{tt} \mathcal{X}(\cdot, s)(t_{n-1} - s) ds,$$

we obtain the following estimate for the temporal consistency error:

$$\Delta t |\mathbf{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \mathbf{e}^n)| = \left| \mathbf{D} \left(\int_{t_{n-1}}^{t_n} \partial_{tt} \mathcal{X}(\cdot, s)(t_{n-1} - s) ds, \mathbf{e}^n \right) \right| \leq \Delta t^{3/2} \left(\int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 ds \right)^{1/2} \|\mathbf{e}^n\|_{\mathbb{D}}.$$

For the nonlinear term, the continuity and skew-symmetry properties of $\mathbf{B}(\cdot, \cdot, \cdot)$ imply

$$\begin{aligned} |\mathbf{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \mathbf{e}^n) - \mathbf{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathbf{e}^n)| &\lesssim \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^{1/2} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}}^{1/2} \|\mathcal{X}(\cdot, t_n) - \mathcal{X}^{n-1}\|_{\mathbb{V}} \|\mathbf{e}^n\|_{\mathbb{D}} \\ &\leq \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^{1/2} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}}^{1/2} \left(\Delta t^{1/2} \left(\int_{t_{n-1}}^{t_n} \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{V}}^2 ds \right)^{1/2} + \|\mathcal{X}^{n-1}\|_{\mathbb{V}} \right) \|\mathbf{e}^n\|_{\mathbb{D}}. \end{aligned}$$

Combining these estimates and applying Young's inequality, we obtain

$$\begin{aligned}
\|\mathbf{e}^n\|_{\mathbb{D}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\mathbf{e}^n\|_{\mathbb{V}}^2 &\leq C\Delta t^{3/2} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^{1/2} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}}^{1/2} \left(\int_{t_{n-1}}^{t_n} \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{V}}^2 ds \right)^{1/2} \|\mathbf{e}^n\|_{\mathbb{D}} \\
&\quad + C\Delta t^{3/2} \left(\int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 ds \right)^{1/2} \|\mathbf{e}^n\|_{\mathbb{D}} + C\Delta t \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^{1/2} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}}^{1/2} \|\mathbf{e}^{n-1}\|_{\mathbb{V}} \|\mathbf{e}^n\|_{\mathbb{D}} \\
&\leq C\Delta t^2 \left(\int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 ds + \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}} \int_{t_{n-1}}^{t_n} \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{V}}^2 ds \right) \\
&\quad + C\Delta t \|\mathbf{e}^n\|_{\mathbb{D}}^2 + \frac{\Delta t}{2} \|\mathbf{e}^{n-1}\|_{\mathbb{V}}^2.
\end{aligned}$$

Summing the above inequality from $n = 1$ to L , for any $1 \leq L \leq N$, and using $\mathbf{e}^0 = 0$, we obtain

$$\|\mathbf{e}^L\|_{\mathbb{D}}^2 + \Delta t \sum_{n=1}^L \|\mathbf{e}^n\|_{\mathbb{V}}^2 \lesssim \Delta t^2 \int_0^{t_L} (\|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{V}}^2) ds + C\Delta t \sum_{n=1}^L \|\mathbf{e}^n\|_{\mathbb{D}}^2.$$

The estimate (36) then follows from the discrete Grönwall inequality.

Next, taking $\mathcal{Y} = \delta_t \mathbf{e}^n \in \mathbb{V}$ in Eq. (39) and applying the Cauchy–Schwarz and Young inequalities, we obtain

$$\|\mathbf{e}^n\|_{\mathbb{A}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{A}}^2 + 2\Delta t \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 \leq 2\Delta t |\mathbb{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \delta_t \mathbf{e}^n)| + 2\Delta t |\mathbb{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \delta_t \mathbf{e}^n) - \mathbb{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \delta_t \mathbf{e}^n)|$$

Arguing as above, we obtain the following bounds:

$$\Delta t |\mathbb{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \delta_t \mathbf{e}^n)| \leq \frac{\Delta t^2}{2} \int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 ds + \frac{\Delta t}{2} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2$$

and

$$\begin{aligned}
&\Delta t |\mathbb{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \delta_t \mathbf{e}^n) - \mathbb{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \delta_t \mathbf{e}^n)| \\
&= \Delta t |\mathbb{B}(\mathcal{X}(\cdot, t_n), \mathbf{e}^n, \delta_t \mathbf{e}^n) + \mathbb{B}(\mathcal{X}(\cdot, t_n) - \mathcal{X}(\cdot, t_{n-1}), \mathcal{X}^n, \delta_t \mathbf{e}^n) + \mathbb{B}(\mathbf{e}^{n-1}, \mathcal{X}^n, \delta_t \mathbf{e}^n)| \\
&\leq C\Delta t \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^{1/2} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}}^{1/2} \|\mathbf{e}^n\|_{\mathbb{V}} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} + C\Delta t^{3/2} \left(\int_{t_{n-1}}^{t_n} \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 ds \right)^{1/2} \|\mathcal{X}^n\|_{\mathbb{V}} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} \\
&\quad + C\Delta t \|\mathcal{X}^n\|_{\mathbb{W}}^{1/2} \|\mathcal{X}^n\|_{\mathbb{V}}^{1/2} \|\mathbf{e}^{n-1}\|_{\mathbb{V}} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} \\
&\leq \frac{\Delta t}{2} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 + C\Delta t \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}} \|\mathbf{e}^n\|_{\mathbb{V}}^2 + C\Delta t^2 \|\mathcal{X}^n\|_{\mathbb{V}}^2 \int_{t_{n-1}}^{t_n} \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 ds \\
&\quad + C\Delta t \|\mathcal{X}^n\|_{\mathbb{W}} \|\mathcal{X}^n\|_{\mathbb{V}} \|\mathbf{e}^{n-1}\|_{\mathbb{V}}^2
\end{aligned}$$

Combining the above estimates yields

$$\begin{aligned}
\|\mathbf{e}^n\|_{\mathbb{A}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{A}}^2 + \Delta t \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 &\leq C\Delta t^2 \int_{t_{n-1}}^{t_n} (\|\partial_{tt} \mathcal{X}(\cdot, t_n)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^2) ds + C\Delta t \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{W}} \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}} \|\mathbf{e}^n\|_{\mathbb{V}}^2 \\
&\quad + C\Delta t \|\mathcal{X}^n\|_{\mathbb{W}} \|\mathcal{X}^n\|_{\mathbb{V}} \|\mathbf{e}^{n-1}\|_{\mathbb{V}}^2
\end{aligned}$$

Summing the above inequality from $n = 1$ to L for any $1 \leq L \leq N$, we obtain

$$\begin{aligned}
\|\mathbf{e}^L\|_{\mathbb{A}}^2 + \Delta t \sum_{n=1}^L \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 &\leq C\Delta t^2 \int_0^T (\|\partial_{tt} \mathcal{X}(\cdot, t_n)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^2) ds + C\Delta t \sum_{n=1}^L \|\mathbf{e}^n\|_{\mathbb{V}}^2 \\
&\leq C\Delta t^2 \int_0^T (\|\partial_{tt} \mathcal{X}(\cdot, t_n)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, t_n)\|_{\mathbb{W}}^2) ds.
\end{aligned}$$

Since L is arbitrary, (37) follows.

The inf-sup condition (28) and the first equation of Eq. (39) imply

$$\begin{aligned}
\|e_p^n\| &\lesssim \sup_{\mathcal{Y} \in \mathbb{V} \setminus \{0\}} \frac{\mathcal{C}(\mathcal{Y}, e_p^n)}{\|\mathcal{Y}\|_{\mathbb{V}}} \\
&= \sup_{\mathcal{Y} \in \mathbb{V} \setminus \{0\}} \frac{\mathbb{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_n), \mathcal{Y}) - [\mathbb{B}(\mathcal{X}(\cdot, t_n), \mathcal{X}(\cdot, t_n), \mathcal{Y}) - \mathbb{B}(\mathcal{X}^{n-1}, \mathcal{X}^n, \mathcal{Y})] - \mathbb{D}(\delta_t \mathbf{e}^n, \mathcal{Y}) - \mathbf{A}(\mathbf{e}^n, \mathcal{Y})}{\|\mathcal{Y}\|_{\mathbb{V}}} \\
&\lesssim \Delta t^{1/2} \left(\int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, t_n)\|_{\mathbb{D}}^2 \right)^{1/2} + \Delta t^{1/2} \left(\int_{t_{n-1}}^{t_n} \|\partial_t \mathcal{X}(\cdot, t_n)\|_{\mathbb{V}}^2 \right)^{1/2} \|\mathcal{X}^n\|_{\mathbb{V}} + \|\mathcal{X}(\cdot, t_n)\|_{\mathbb{V}} \|\mathbf{e}^n\|_{\mathbb{V}} \\
&\quad + \|\mathcal{X}^n\|_{\mathbb{V}} \|\mathbf{e}^{n-1}\|_{\mathbb{V}} + \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} + \|\mathbf{e}^n\|_{\mathbb{V}}.
\end{aligned}$$

Therefore,

$$\Delta t \sum_{n=1}^N \|e_p^n\|^2 \lesssim \Delta t^2 \int_0^T (\|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_t \mathcal{X}(\cdot, s)\|_{\mathbb{V}}^2) ds + \Delta t \sum_{n=1}^N \|\mathbf{e}^n\|_{\mathbb{V}}^2 + \Delta t \sum_{n=1}^N \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2.$$

Combining this estimate with Eqs. (36) and (37) yields (38). $\square$

4.2 Spatial discretization

For $\mathcal{X}_h = (\mathbf{u}_h, \boldsymbol{\omega}_h, \mathbf{m}_h, \varphi_h)^\top$, $\mathcal{Y}_h = (\mathbf{v}_h, \mathbf{s}_h, \mathbf{F}_h, \psi_h)^\top$, and $\mathcal{Z}_h = (\boldsymbol{\epsilon}_h, \boldsymbol{\sigma}_h, \mathbf{A}_h, \phi_h)^\top$ in $\mathbb{V}_h^1$, we define the bilinear form $\mathbf{A}_h^1 : \mathbb{V}_h^1 \times \mathbb{V}_h^1 \rightarrow \mathbb{R}$ by

$$\begin{aligned}
\mathbf{A}_h^1(\mathcal{X}_h, \mathcal{Y}_h) &:= \eta(\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) + \eta'(\nabla \boldsymbol{\omega}_h, \nabla \mathbf{s}_h) + (\eta' + \lambda')(\operatorname{div} \boldsymbol{\omega}_h, \operatorname{div} \mathbf{s}_h) + \sigma(\operatorname{curl}_h^1 \mathbf{m}_h, \operatorname{curl}_h^1 \mathbf{F}_h) + \sigma(\operatorname{div} \mathbf{m}_h, \operatorname{div} \mathbf{F}_h) \\
&\quad + \frac{1}{\tau}(\mathbf{m}_h, \mathbf{F}_h) + \frac{\mu_0(1 + \chi_0) + \chi_0}{\tau}(\nabla \varphi_h, \nabla \psi_h) + \sigma \mu_0(\Delta_h^1 \varphi_h, \Delta_h^1 \psi_h) + \zeta(\operatorname{curl} \mathbf{u}_h, \operatorname{curl} \mathbf{v}_h) - 2\zeta(\boldsymbol{\omega}_h, \operatorname{curl} \mathbf{v}_h) \\
&\quad - 2\zeta(\operatorname{curl} \mathbf{u}_h, \mathbf{s}_h) + 4\zeta(\boldsymbol{\omega}_h, \mathbf{s}_h),
\end{aligned}$$

and the trilinear form $\mathbf{B}_h^1 : \mathbb{V}_h^1 \times \mathbb{V}_h^1 \times \mathbb{V}_h^1 \rightarrow \mathbb{R}$ by

$$\begin{aligned}
\mathbf{B}_h^1(\mathcal{Z}_h, \mathcal{X}_h, \mathcal{Y}_h) &:= \rho b(\boldsymbol{\epsilon}_h, \mathbf{u}_h, \mathbf{v}_h) + \rho \kappa b(\boldsymbol{\epsilon}_h, \boldsymbol{\omega}_h, \mathbf{s}_h) - c(\boldsymbol{\epsilon}_h, \mathbf{m}_h, \mathbf{F}_h) - \frac{1}{2}(\mathbf{m}_h \times \operatorname{curl} \boldsymbol{\epsilon}_h, \mathbf{F}_h) - (\boldsymbol{\sigma}_h \times \mathbf{m}_h, \mathbf{F}_h) \\
&\quad + \mu_0[(\nabla \varphi_h, \mathbf{v}_h \operatorname{div} \mathbf{A}_h) - (\nabla \psi_h, \mathbf{u}_h \operatorname{div} \mathbf{A}_h)] + \frac{\mu_0}{2}[\mathbf{u}_h \times \operatorname{curl}_h^1 \mathbf{A}_h, \nabla \psi_h] - (\mathbf{v}_h \times \operatorname{curl}_h^1 \mathbf{A}_h, \nabla \varphi_h) \\
&\quad + \frac{\mu_0}{2}[(\mathbf{A}_h \times \operatorname{curl} \mathbf{u}_h, \nabla \psi_h) - (\mathbf{A}_h \times \operatorname{curl} \mathbf{v}_h, \nabla \varphi_h)] + \mu_0[(\boldsymbol{\omega}_h \times \mathbf{A}_h, \nabla \psi_h) - (\mathbf{s}_h \times \mathbf{A}_h, \nabla \varphi_h)] \\
&\quad + \frac{\chi_0}{\tau}[(\mathbf{m}_h, \nabla \psi_h) - (\mathbf{F}_h, \nabla \varphi_h)] + \frac{1}{2}[(\boldsymbol{\epsilon}_h \times \mathbf{F}_h, \operatorname{curl}_h^1 \mathbf{m}_h) - (\boldsymbol{\epsilon}_h \times \mathbf{m}_h, \operatorname{curl}_h^1 \mathbf{F}_h)] \\
&\quad + \frac{\mu_0}{2}[(\Delta_h^1 \psi_h, \mathbf{u}_h \cdot \mathbf{A}_h) - (\Delta_h^1 \varphi_h, \mathbf{v}_h \cdot \mathbf{A}_h)].
\end{aligned}$$

The definitions of these forms immediately yield the following structural properties.

Lemma 4.6. *The bilinear form $\mathbf{A}_h^1$ and the trilinear form $\mathbf{B}_h^1$ satisfy the following properties:*

(1) *The bilinear form $\mathbf{A}_h^1$ is bounded, symmetric, and coercive; namely,*

$$\mathbf{A}_h^1(\mathcal{X}_h, \mathcal{Y}_h) \lesssim \|\mathcal{X}_h\|_{\mathbb{V}_h^1} \|\mathcal{Y}_h\|_{\mathbb{V}_h^1} \quad \forall \mathcal{X}_h, \mathcal{Y}_h \in \mathbb{V}_h^1, \quad (40)$$

$$\mathbf{A}_h^1(\mathcal{X}_h, \mathcal{Y}_h) = \mathbf{A}_h^1(\mathcal{Y}_h, \mathcal{X}_h) \quad \forall \mathcal{X}_h, \mathcal{Y}_h \in \mathbb{V}_h^1, \quad (41)$$

$$\mathbf{A}_h^1(\mathcal{Y}_h, \mathcal{Y}_h) \gtrsim \|\mathcal{Y}_h\|_{\mathbb{V}_h^1}^2 \quad \forall \mathcal{Y}_h \in \mathbb{V}_h^1. \quad (42)$$

(2) *The trilinear form $\mathbf{B}_h^1$ is skew-symmetric with respect to its second and third arguments; namely,*

$$\mathbf{B}_h^1(\mathcal{Z}_h, \mathcal{X}_h, \mathcal{Y}_h) = -\mathbf{B}_h^1(\mathcal{Z}_h, \mathcal{Y}_h, \mathcal{X}_h) \quad \forall \mathcal{X}_h, \mathcal{Y}_h, \mathcal{Z}_h \in \mathbb{V}_h^1, \quad (43)$$

Moreover, it satisfies the continuity estimate

$$\mathbf{B}_h^1(\mathcal{Z}_h, \mathcal{X}_h, \mathcal{Y}_h) \lesssim \|\mathcal{Z}_h\|_{\mathbb{V}_h^1} \|\mathcal{X}_h\|_{\mathbb{V}_h^1} \|\mathcal{Y}_h\|_{\mathbb{V}_h^1} \quad \forall \mathcal{Z}_h, \mathcal{X}_h, \mathcal{Y}_h \in \mathbb{V}_h^1. \quad (44)$$

For each $1 \leq n \leq N$, we define the following first-order fully discrete scheme for (30).

Algorithm 2 First-order fully discrete scheme

Given $\mathcal{X}_h^{n-1} \in \mathbb{V}_h^1$, find $\mathcal{X}_h^n \in \mathbb{V}_h^1$ and $\tilde{p}^n \in L_h$ such that

$$\begin{cases} \mathcal{D}(\delta_t \mathcal{X}_h^n, \mathcal{Y}_h) + \mathbf{A}_h^1(\mathcal{X}_h^n, \mathcal{Y}_h) + \mathbf{B}_h^1(\mathcal{X}_h^{n-1}, \mathcal{X}_h^n, \mathcal{Y}_h) + \mathcal{C}(\mathcal{Y}_h, \tilde{p}_h^n) = (\mathbf{f}(\cdot, t_n), \mathbf{v}_h), & \forall \mathcal{Y}_h \in \mathbb{V}_h^1, \\ \mathcal{C}(\mathcal{X}_h^n, q_h) = 0, & \forall q_h \in L_h, \end{cases} \quad (45)$$

with initial condition

$$\mathcal{X}_h^0 = \mathcal{I}_h^1 \mathcal{X}_0. \quad (46)$$

At each time level t_n , for any discrete quantity $\mathbf{v}_h$, we use the notation

$$\mathbf{v}_h^- := \mathbf{v}_h^{n-1}, \quad \mathbf{v}_h := \mathbf{v}_h^n, \quad \delta_t \mathbf{v}_h := \delta_t \mathbf{v}_h^n.$$

With this notation, the fully discrete scheme (45) in Algorithm 2 can be written equivalently in the following computable form. Find

$$\begin{aligned} \mathbf{u}_h &\in \mathbf{S}_h^1, & \tilde{p}_h &\in L_h, & \boldsymbol{\omega}_h &\in \boldsymbol{\Sigma}_h^1, & \mathbf{m}_h &\in \mathbf{V}_h^1, \\ \mathbf{z}_h &\in \mathbf{U}_h^1, & \mathbf{k}_h &\in \mathbf{U}_h^1, & \boldsymbol{\varphi}_h &\in \boldsymbol{\Sigma}_h^1, & \boldsymbol{\vartheta}_h &\in \boldsymbol{\Sigma}_h^1, & \boldsymbol{\varpi}_h &\in \boldsymbol{\Sigma}_h^1, \end{aligned}$$

such that, for all

$$\begin{aligned} \mathbf{v}_h &\in \mathbf{S}_h^1, & \tilde{q}_h &\in L_h, & \mathbf{s}_h &\in \boldsymbol{\Sigma}_h^1, & \mathbf{F}_h &\in \mathbf{V}_h^1, \\ \boldsymbol{\lambda}_h &\in \mathbf{U}_h^1, & \boldsymbol{\theta}_h &\in \mathbf{U}_h^1, & \boldsymbol{\psi}_h &\in \boldsymbol{\Sigma}_h^1, & \varsigma_h &\in \boldsymbol{\Sigma}_h^1, & \boldsymbol{\gamma}_h &\in \boldsymbol{\Sigma}_h^1 \end{aligned}$$

such that

$$\left\{ \begin{aligned} & \rho(\delta_t \mathbf{u}_h, \mathbf{v}_h) + \rho b(\mathbf{u}_h^-, \mathbf{u}_h, \mathbf{v}_h) + \eta(\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) + \zeta(\text{curl } \mathbf{u}_h, \text{curl } \mathbf{v}_h) - 2\zeta(\boldsymbol{\omega}_h, \text{curl } \mathbf{v}_h) - (\tilde{p}_h, \text{div } \mathbf{v}_h) \\ & \quad + \frac{\mu_0}{2}(\boldsymbol{\vartheta}_h, \mathbf{m}_h^- \cdot \mathbf{v}_h) + \frac{\mu_0}{2}(\nabla \boldsymbol{\varphi}_h \text{div } \mathbf{m}_h^-, \mathbf{v}_h) - \frac{\mu_0}{2}(\mathbf{v}_h \times \mathbf{k}_h^-, \nabla \boldsymbol{\varphi}_h) - \frac{\mu_0}{2}(\mathbf{m}_h^- \times \text{curl } \mathbf{v}_h, \nabla \boldsymbol{\varphi}_h) = 0, \\ & \quad (\text{div } \mathbf{u}_h, q_h) = 0, \\ & \rho \kappa(\delta_t \boldsymbol{\omega}_h, \mathbf{s}_h) + \eta'(\nabla \boldsymbol{\omega}_h, \nabla \mathbf{s}_h) + (\eta' + \lambda')(\text{div } \boldsymbol{\omega}_h, \text{div } \mathbf{s}_h) + 4\zeta(\boldsymbol{\omega}_h, \mathbf{s}_h) + \rho \kappa b(\mathbf{u}_h^-, \boldsymbol{\omega}_h, \mathbf{s}_h) \\ & \quad - \mu_0(\mathbf{m}_h^- \times \nabla \boldsymbol{\varphi}_h, \mathbf{s}_h) - 2\zeta(\text{curl } \mathbf{u}_h, \mathbf{s}_h) = 0, \\ & (\delta_t \mathbf{m}_h, \mathbf{F}_h) + \frac{1}{\tau}(\mathbf{m}_h, \mathbf{F}_h) + \sigma(\text{curl } \mathbf{k}_h, \mathbf{F}_h) + \sigma(\text{div } \mathbf{m}_h, \text{div } \mathbf{F}_h) - c(\mathbf{u}_h^-, \mathbf{m}_h, \mathbf{F}_h) - \frac{1}{2}(\mathbf{m}_h \times \text{curl } \mathbf{u}_h^-, \mathbf{F}_h) \\ & \quad - \frac{1}{2}(\mathbf{u}_h^- \times \mathbf{k}_h, \mathbf{F}_h) - \frac{1}{2}(\text{curl } \mathbf{z}_h, \mathbf{F}_h) - (\boldsymbol{\omega}_h^- \times \mathbf{m}_h, \mathbf{F}_h) - \frac{\chi_0}{\tau}(\nabla \boldsymbol{\varphi}_h, \mathbf{F}_h) = 0, \\ & \quad (\mathbf{z}_h, \boldsymbol{\lambda}_h) - (\mathbf{u}_h^- \times \mathbf{m}_h, \boldsymbol{\lambda}_h) = 0, \\ & \quad (\mathbf{k}_h, \boldsymbol{\theta}_h) - (\mathbf{m}_h, \text{curl } \boldsymbol{\theta}_h) = 0, \\ & \mu_0(\delta_t \nabla \boldsymbol{\varphi}_h, \nabla \boldsymbol{\psi}_h) + \frac{\mu_0(1+\chi_0)+\chi_0}{\tau}(\nabla \boldsymbol{\varphi}_h, \nabla \boldsymbol{\psi}_h) + \sigma \mu_0(\nabla \boldsymbol{\vartheta}_h, \nabla \boldsymbol{\psi}_h) + \mu_0(\boldsymbol{\omega}_h \times \mathbf{m}_h^-, \nabla \boldsymbol{\psi}_h) + \frac{\chi_0}{\tau}(\mathbf{m}_h, \nabla \boldsymbol{\psi}_h) \\ & \quad - \frac{\mu_0}{2}(\nabla \boldsymbol{\varpi}_h, \nabla \boldsymbol{\psi}_h) - \frac{\mu_0}{2}(\mathbf{u}_h \text{div } \mathbf{m}_h^-, \nabla \boldsymbol{\psi}_h) + \frac{\mu_0}{2}(\mathbf{m}_h^- \times \text{curl } \mathbf{u}_h, \nabla \boldsymbol{\psi}_h) + \frac{\mu_0}{2}(\mathbf{u}_h \times \mathbf{k}_h^-, \nabla \boldsymbol{\psi}_h) \\ & \quad + (\partial_t \mathbf{H}_e, \nabla \boldsymbol{\psi}_h) + \frac{\mu_0+\chi_0}{\tau \mu_0}(\mathbf{H}_e, \nabla \boldsymbol{\psi}_h) + \sigma(\text{grad div } \mathbf{H}_e, \nabla \boldsymbol{\psi}_h) = 0, \\ & \sigma \mu_0(\boldsymbol{\vartheta}_h, \varsigma_h) - \sigma \mu_0(\nabla \boldsymbol{\varphi}_h, \nabla \varsigma_h) = 0, \\ & (\boldsymbol{\varpi}_h, \boldsymbol{\gamma}_h) - (\mathbf{u}_h \cdot \mathbf{m}_h^-, \boldsymbol{\gamma}_h) = 0. \end{aligned} \right. \quad (47)$$

Once (47) has been solved, the discrete magnetic field $\mathbf{H}_h \in \mathbf{U}_h^1$ is recovered from

$$(\mathbf{H}_h, \boldsymbol{\varepsilon}_h) = (\nabla \boldsymbol{\varphi}_h, \boldsymbol{\varepsilon}_h) \quad \forall \boldsymbol{\varepsilon}_h \in \mathbf{U}_h^1. \quad (48)$$

Consequently, $\mathbf{H}_h = \nabla \boldsymbol{\varphi}_h$, and hence $\text{curl } \mathbf{H}_h = 0$ pointwise in Ω .

We emphasize that (47) is linear with respect to the unknown vector $\mathbf{X}_h = (\mathbf{u}_h, \tilde{p}_h, \boldsymbol{\omega}_h, \mathbf{m}_h, \mathbf{z}_h, \mathbf{k}_h, \boldsymbol{\varphi}_h, \boldsymbol{\vartheta}_h)^\top$. Let

$$\mathbf{X}_{h,1} = (\mathbf{u}_h, \tilde{p}_h)^\top, \quad \mathbf{X}_{h,2} = \boldsymbol{\omega}_h, \quad \mathbf{X}_{h,3} = (\mathbf{m}_h, \mathbf{z}_h, \mathbf{k}_h)^\top, \quad \mathbf{X}_{h,4} = (\boldsymbol{\varphi}_h, \boldsymbol{\vartheta}_h)^\top, \quad \mathbf{X}_{h,5} = \boldsymbol{\varpi}_h.$$

With this block decomposition, the linear system (47) can be written as

$$\begin{pmatrix} K_1 & C_{12} & 0 & C_{14} & 0 \\ C_{21} & K_2 & 0 & C_{24} & 0 \\ 0 & 0 & K_3 & C_{34} & 0 \\ C_{41} & C_{42} & C_{43} & K_4 & C_{45} \\ C_{51} & 0 & 0 & 0 & K_5 \end{pmatrix} \begin{pmatrix} \mathbf{X}_{h,1} \\ \mathbf{X}_{h,2} \\ \mathbf{X}_{h,3} \\ \mathbf{X}_{h,4} \\ \mathbf{X}_{h,5} \end{pmatrix} = \begin{pmatrix} \mathbf{F}_1 \\ \mathbf{F}_2 \\ \mathbf{F}_3 \\ \mathbf{F}_4 \\ \mathbf{F}_5 \end{pmatrix},$$

This block system can be solved efficiently by the exact Schur complement decoupling solver described in Algorithm 3.

By arguments analogous to those used in the proofs of Theorems 4.1 and 4.2, we obtain the following well-posedness and energy-stability result for Algorithm 2.

Algorithm 3 Exact Schur complement decoupling solver**Step 1. Solve the velocity-pressure block.**

Solve

$$K_1 \mathbf{X}_1 = \mathbf{F}_1, \quad K_1 Y_{12} = C_{12}, \quad K_1 Y_{14} = C_{14}.$$

Step 2. Solve the angular velocity block.

Define

$$S_2 = K_2 - C_{21} Y_{12}.$$

Solve

$$S_2 \mathbf{X}_2 = \mathbf{F}_2 - C_{21} \mathbf{X}_1,$$

and

$$S_2 Y_{24} = C_{24} - C_{21} Y_{14}.$$

Step 3. Solve the magnetization block.

Solve

$$K_3 \mathbf{X}_3 = \mathbf{F}_3, \quad K_3 Y_{34} = C_{34}.$$

Step 4. Solve the auxiliary scalar block.

Set

$$\mathbf{X}_0 = \mathbf{X}_1 - Y_{12} \mathbf{X}_2,$$

and

$$T_1 = Y_{12} Y_{24} - Y_{14}.$$

Solve

$$K_5 \mathbf{X}_5 = \mathbf{F}_5 - C_{51} \mathbf{X}_0,$$

and

$$K_5 Y_{54} = C_{51} T_1.$$

Step 5. Solve the Schur complement system for (φ_h, ϑ_h) .

Define

$$S_4 = K_4 + C_{41} T_1 - C_{42} Y_{24} - C_{43} Y_{34} - C_{45} Y_{54}.$$

Solve

$$S_4 \mathbf{X}_{h,4} = \mathbf{F}_4 - C_{41} \mathbf{X}_0 - C_{42} \mathbf{X}_2 - C_{43} \mathbf{X}_3 - C_{45} \mathbf{X}_5.$$

Step 6. Recover the remaining variables.

Set

$$\mathbf{X}_{h,2} = \mathbf{X}_2 - Y_{24} \mathbf{X}_{h,4},$$

$$\mathbf{X}_{h,1} = \mathbf{X}_1 - Y_{12} \mathbf{X}_{h,2} - Y_{14} \mathbf{X}_{h,4},$$

$$\mathbf{X}_{h,3} = \mathbf{X}_3 - Y_{34} \mathbf{X}_{h,4},$$

$$\mathbf{X}_{h,5} = \mathbf{X}_5 - Y_{54} \mathbf{X}_{h,4}.$$

Step 7. Output

$$(\mathbf{u}_h, \tilde{p}_h), \quad \omega_h, \quad (\mathbf{m}_h, \mathbf{z}_h, \mathbf{k}_h), \quad (\varphi_h, \vartheta_h), \quad \varpi_h.$$

Theorem 4.7 (Well-posedness and energy estimate for Algorithm 2). *Given $\mathcal{X}_h^{n-1} \in \mathbb{V}_h^1$ for $1 \leq n \leq N$, the fully discrete scheme Algorithm 2 admits a unique solution $(\mathcal{X}_h^n, \tilde{p}_h^n) \in \mathbb{V}_h^1 \times L_h$ at each time step. Moreover, the solution sequence $\{(\mathcal{X}_h^n, \tilde{p}_h^n)\}_{n=1}^N$ satisfies the following energy estimate*

$$\left(\max_{1 \leq n \leq N} \|\mathcal{X}_h^n\|_{\mathbb{D}}^2 + \Delta t \sum_{n=1}^N \|\mathcal{X}_h^n\|_{\mathbb{A}_h^1}^2 \right)^{1/2} \leq \|\mathcal{X}_h^0\|_{\mathbb{D}} + 2\Delta t \sum_{n=1}^N \|\mathbf{f}(\cdot, t_n)\|_{\mathbb{D}^{-1}}. \quad (49)$$

The fully discrete error analysis can be carried out along the same lines as that of the time semi-discrete scheme by introducing suitable quasi-interpolation operators for the spatial discretization. The main additional issue is the consistency error associated with the discrete differential operators, which can be treated by standard techniques from the error analysis of nonconforming finite element approximations. Since the resulting arguments are largely technical and do not introduce essentially new ideas, we omit the details here for brevity. A complete derivation of the fully discrete error estimates is available in the arXiv version of this paper.

5 Second-Order Scheme

This section presents the construction of a second-order discrete scheme for the ferrohydrodynamic equations. We employ a modified Crank-Nicolson method for time discretization and adopt the finite element exterior calculus framework for spatial discretization.

5.1 Time discretization

In this subsection, we develop a second-order accurate time discretization for the ferrohydrodynamic equations. Building upon the abstract formulation, the scheme is designed to achieve higher temporal accuracy while preserving the intrinsic energy stability of the continuous problem.

Algorithm 4 Second Order Scheme

Given $\mathcal{X}^{n-1} \in \mathbb{V}$, find $\mathcal{X}^n \in \mathbb{V}$ and $\tilde{p}^n \in W$ such that

$$\begin{cases} \mathbb{D}(\delta_t \mathcal{X}^n, \mathcal{Y}) + \mathbb{A}(\bar{\mathcal{X}}^{n-1/2}, \mathcal{Y}) + \mathbb{B}(\hat{\mathcal{X}}^{n-1}, \bar{\mathcal{X}}^{n-1/2}, \mathcal{Y}) + \mathbb{C}(\mathcal{Y}, \tilde{p}^n) = (\mathbf{f}(\cdot, t_{n-1/2}), \mathcal{Y}) & \forall \mathcal{Y} \in \mathbb{V}, \\ \mathbb{C}(\bar{\mathcal{X}}^{n-1/2}, q) = 0 & \forall q \in W, \end{cases} \quad (50)$$

with initial condition

$$\mathcal{X}^0 = \mathcal{X}_0.$$

Remark 2. In Algorithm 4, the solution at the first time level is computed using the fully nonlinear scheme to ensure consistency with the original problem and to provide an accurate starting value. For all subsequent time steps, a linearized implicit scheme is employed, in which the first argument of the trilinear form $\mathbb{B}(\cdot, \cdot, \cdot)$ is treated explicitly. This strategy reduces computational cost while preserving stability and the overall second-order temporal accuracy of the method.

Similar to the analysis of Algorithm 1, we obtain the following existence and uniqueness result for the semi-discrete scheme Algorithm 4.

Theorem 5.1. *For any $2 \leq n \leq N$, given $\mathcal{X}^{n-1} \in \mathbb{V}$ and $\mathcal{X}^{n-2} \in \mathbb{V}$, the semi-discrete scheme Algorithm 4 admits a unique solution $(\mathcal{X}^n, \tilde{p}^n) \in \mathbb{V} \times W$ at each time step.*

To investigate the stability of the scheme, we derive discrete energy estimates. By selecting appropriate test functions and exploiting the structural properties of the formulation, we obtain bounds that reflect the dissipative nature of the system.

Theorem 5.2 (Energy estimate for Algorithm 4). *Given $\mathcal{X}^{n-1} \in \mathbb{V}$ for $1 \leq n \leq N$, let $(\mathcal{X}^n, \tilde{p}^n) \in \mathbb{V} \times W$ be the unique solution of Algorithm 4. Then the following energy estimate holds:*

$$\left(\max_{0 \leq n \leq N} \|\mathcal{X}^n\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^N \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{A}}^2 \right)^{1/2} \leq \|\mathcal{X}^0\|_{\mathbb{D}} + 2\Delta t \sum_{n=1}^N \|\mathbf{f}(\cdot, t_{n-1/2})\|_{\mathbb{D}^{-1}}. \quad (51)$$

Proof. The second equation of (50) implies that $\bar{\mathcal{X}}^{n-1/2} \in \mathfrak{K}$. Taking $\mathcal{Y} = \bar{\mathcal{X}}^{n-1/2} \in \mathfrak{K} \subset \mathbb{V}$ as a test function in the first equation of (50), we obtain

$$\|\mathcal{X}^n\|_{\mathbb{D}}^2 - \|\mathcal{X}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{A}}^2 = 2\Delta t (\mathbf{f}(\cdot, t_{n-1/2}), \bar{\mathcal{X}}^{n-1/2}).$$

Applying the Cauchy–Schwarz inequality yields

$$2\Delta t(\mathbf{f}(\cdot, t_{n-1/2}), \bar{\mathcal{X}}^{n-1/2}) \leq \Delta t \|\mathbf{f}(\cdot, t_{n-1/2})\|_{\mathbb{D}^{-1}} (\|\mathcal{X}^n\|_{\mathbb{D}} + \|\mathcal{X}^{n-1}\|_{\mathbb{D}}).$$

Hence,

$$\|\mathcal{X}^n\|_{\mathbb{D}}^2 - \|\mathcal{X}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{A}}^2 \leq \Delta t \|\mathbf{f}(\cdot, t_{n-1/2})\|_{\mathbb{D}^{-1}} (\|\mathcal{X}^n\|_{\mathbb{D}} + \|\mathcal{X}^{n-1}\|_{\mathbb{D}}).$$

Summing from $n = 1$ to L for any $1 \leq L \leq N$, we obtain

$$\|\mathcal{X}^L\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^L \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{A}}^2 \leq \|\mathcal{X}^0\|_{\mathbb{D}}^2 + \Delta t \sum_{n=1}^L \|\mathbf{f}(\cdot, t_{n-1/2})\|_{\mathbb{D}^{-1}} (\|\mathcal{X}^n\|_{\mathbb{D}} + \|\mathcal{X}^{n-1}\|_{\mathbb{D}}).$$

Taking the maximum over $0 \leq n \leq N$ and using the arbitrariness of L , we deduce

$$\max_{0 \leq n \leq N} \|\mathcal{X}^n\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^N \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{A}}^2 \leq \|\mathcal{X}^0\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^N \|\mathbf{f}(\cdot, t_{n-1/2})\|_{\mathbb{D}^{-1}} \max_{0 \leq n \leq N} \|\mathcal{X}^n\|_{\mathbb{D}}.$$

The desired estimate then follows from Lemma 3.4. $\square$

Remark 3. It follows from the proof of Theorem 5.2 that, in the absence of the external field (i.e., $\mathbf{H}_e = 0$), the scheme Algorithm 4 preserves the total energy in the sense that

$$\|\mathcal{X}^L\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^L \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{A}}^2 = \|\mathcal{X}^0\|_{\mathbb{D}}^2 \quad \text{for any } 1 \leq L \leq N.$$

In what follows, we focus on the error analysis of Algorithm 4. Analogously to the analysis for Algorithm 1, we begin by rewriting the weak formulation (30) at the intermediate time level $t_{n-1/2}$ ($1 \leq n \leq N$) in an equivalent form:

$$\begin{cases} \mathbb{D}(\delta_t \mathcal{X}(\cdot, t_n), \mathcal{Y}) + \mathbb{A}(\bar{\mathcal{X}}(\cdot, t_{n-1/2}), \mathcal{Y}) + \mathbb{B}(\mathcal{X}(\cdot, t_{n-1/2}), \mathcal{X}(\cdot, t_{n-1/2}), \mathcal{Y}) + \mathbb{C}(\mathcal{Y}, \bar{p}(\cdot, t_{n-1/2})) \\ \quad = (\mathbf{f}(\cdot, t_{n-1/2}), \mathcal{Y}) + R_1(\mathcal{Y}) + R_2(\mathcal{Y}) + R_3(\mathcal{Y}), \quad \forall \mathcal{Y} \in \mathbb{V}, \\ \mathbb{C}(\bar{\mathcal{X}}(\cdot, t_{n-1/2}), q) = 0, \quad \forall q \in W, \end{cases} \quad (52)$$

where

$$\begin{aligned} R_1(\mathcal{Y}) &:= \mathbb{D}(\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_{n-1/2}), \mathcal{Y}), \\ R_2(\mathcal{Y}) &:= \mathbb{A}(\bar{\mathcal{X}}(\cdot, t_{n-1/2}) - \mathcal{X}(\cdot, t_{n-1/2}), \mathcal{Y}), \\ R_3(\mathcal{Y}) &:= \mathbb{C}(\mathcal{Y}, \bar{p}(\cdot, t_{n-1/2}) - \tilde{p}(\cdot, t_{n-1/2})). \end{aligned}$$

Let $\mathbf{e}^n = \mathcal{X}(\cdot, t_n) - \mathcal{X}^n$ and $e_p^n = \bar{p}(\cdot, t_n) - \tilde{p}^n$. We are now in a position to derive the corresponding error estimates.

Theorem 5.3 (Error estimates for Algorithm 4). *Let $(\mathcal{X}, \bar{p}) : (0, T] \rightarrow \mathbb{V} \times W$ be the solution of the continuous weak formulation (30), and let $\{(\mathcal{X}^n, \tilde{p}^n)\}_{n=1}^N \subset \mathbb{V} \times W$ be the unique solution of the semi-discrete scheme Algorithm 4. Then the following error estimates hold:*

$$\begin{aligned} & \max_{1 \leq n \leq N} \|\mathcal{X}(\cdot, t_n) - \mathcal{X}^n\|_{\mathbb{D}}^2 + \Delta t \sum_{n=1}^N \|\bar{\mathcal{X}}(\cdot, t_{n-1/2}) - \bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{A}}^2 \\ & \leq C\Delta t^4 \int_0^T \left(\|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 \right) ds, \end{aligned} \quad (53)$$

$$\begin{aligned} & \max_{1 \leq n \leq N} \|\mathcal{X}(\cdot, t_n) - \mathcal{X}^n\|_{\mathbb{A}}^2 + \Delta t \sum_{n=1}^N \|\delta_t(\mathcal{X}(\cdot, t_n) - \mathcal{X}^n)\|_{\mathbb{D}}^2 \\ & \leq C\Delta t^4 \int_0^T \left(\|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 \right) ds, \end{aligned} \quad (54)$$

$$\begin{aligned} & \Delta t \sum_{n=1}^N \|\bar{p}(\cdot, t_{n-1/2}) - \tilde{p}^{n-1/2}\|^2 \\ & \leq C\Delta t^4 \int_0^T \left(\|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 + \|\partial_{tt} \tilde{p}(\cdot, s)\|^2 + \|\partial_{ttt} \tilde{p}(\cdot, s)\|^2 \right) ds. \end{aligned} \quad (55)$$

Proof. Subtracting Algorithm 4 from (52), we obtain

$$\begin{cases} \mathbf{D}(\delta_t \mathbf{e}^n, \mathcal{Y}) + \mathbf{A}(\bar{\mathbf{e}}^{n-1/2}, \mathcal{Y}) + \mathbf{C}(\mathcal{Y}, \bar{e}_p^{n-1/2}) = R_1(\mathcal{Y}) + R_2(\mathcal{Y}) + R_3(\mathcal{Y}) \\ \quad - [\mathbf{B}(\mathcal{X}(\cdot, t_{n-1/2}), \mathcal{X}(\cdot, t_{n-1/2}), \mathcal{Y}) - \mathbf{B}(\hat{\mathcal{X}}^{n-1}, \bar{\mathcal{X}}^{n-1/2}, \mathcal{Y})] \quad \forall \mathcal{Y} \in \mathbb{V}, \\ \mathbf{C}(\bar{\mathbf{e}}^{n-1/2}, q) = 0 \quad \forall q \in W. \end{cases} \quad (56)$$

Taking the test functions $\mathcal{Y} = \bar{\mathbf{e}}^{n-1/2} \in \mathbb{V}$ and $q = \bar{e}_p^{n-1/2} \in W$ in Eq. (56), we obtain

$$\begin{aligned} \|\mathbf{e}^n\|_{\mathbb{D}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}}^2 &= 2\Delta t R_1(\bar{\mathbf{e}}^{n-1/2}) + 2\Delta t R_2(\bar{\mathbf{e}}^{n-1/2}) \\ &\quad - 2\Delta t [\mathbf{B}(\mathcal{X}(\cdot, t_{n-1/2}), \mathcal{X}(\cdot, t_{n-1/2}), \bar{\mathbf{e}}^{n-1/2}) - \mathbf{B}(\hat{\mathcal{X}}^{n-1}, \bar{\mathcal{X}}^{n-1/2}, \bar{\mathbf{e}}^{n-1/2})] \\ &\leq 2\Delta t \|\delta_t \mathcal{X}(\cdot, t_n) - \partial_t \mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{D}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{D}} + 2\Delta t \|\bar{\mathcal{X}}(\cdot, t_{n-1/2}) - \mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{A}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \\ &\quad + C\Delta t \|\mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\mathcal{X}(\cdot, t_{n-1/2}) - \bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{V}} \\ &\quad + C\Delta t \|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \|\hat{\mathcal{X}}(\cdot, t_{n-1}) - \mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{A}} \\ &\quad + C\Delta t \|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{W}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{V}} \|\hat{\mathbf{e}}^{n-1}\|_{\mathbb{D}}. \end{aligned}$$

For any quantity $J(t)$, using the identities:

$$\begin{aligned} J(t_n) - J(t_{n-1}) - \Delta t \partial_t J(t_{n-1/2}) &= -\frac{1}{2} \int_{t_{n-1}}^{t_n} (s - t_{n-1})(t_n - s) \partial_{ttt} J(s) ds + \frac{\Delta t}{2} \int_{t_{n-1/2}}^{t_n} \int_{s-\Delta t/2}^s \partial_{ttt} J(l) dl ds, \\ \bar{J}(t_{n-1/2}) - J(t_{n-1/2}) &= \frac{1}{2} \int_{t_{n-1}}^{t_{n-1/2}} (t_{n-1} - s) \int_s^{s+\Delta t/2} \partial_{ttt} J(l) dl ds + \frac{\Delta t}{4} \int_{t_{n-1}}^{t_{n-1/2}} \partial_{tt} J(s + \Delta t/2) ds, \\ \hat{J}(t_{n-1}) - J(t_{n-1/2}) &= \begin{cases} -\frac{1}{2} \int_{t_{n-1}}^{t_n} \int_{s-\Delta t}^{s-\Delta t/2} \partial_{tt} J(l) dl ds - \frac{1}{2} \int_{t_{n-1}}^{t_{n-1/2}} \int_{s-\Delta t/2}^s \partial_{tt} J(l) dl ds, & \text{for } n \geq 2, \\ \frac{1}{2} \int_{t_0}^{t_{1/2}} (t_0 - s) \int_s^{s+\Delta t/2} \partial_{ttt} J(l) dl ds + \frac{\Delta t}{4} \int_{t_0}^{t_{1/2}} \partial_{tt} J(s + \Delta t/2) ds, & \text{for } n = 1, \end{cases} \end{aligned}$$

we have

$$\begin{aligned} \|\mathbf{e}^n\|_{\mathbb{D}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}}^2 &\leq C\Delta t^{5/2} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{D}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 ds \right)^{1/2} + C\Delta t^{7/2} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} \\ &\quad + C\Delta t^{5/2} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} + C\Delta t^{7/2} \|\mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} \\ &\quad + C\Delta t^{5/2} \|\mathcal{X}(\cdot, t_{n-1})\|_{\mathbb{V}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} + C\Delta t \|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{W}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \|\hat{\mathbf{e}}^{n-1}\|_{\mathbb{D}} \\ &\quad + \begin{cases} C\Delta t^{5/2} \|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} \left(\int_{t_{n-2}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} & \text{for } n \geq 2, \\ C\|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\bar{\mathbf{e}}^{1/2}\|_{\mathbb{A}} \left(\Delta t^7 \int_{t_0}^{t_1} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds + \Delta t^5 \int_{t_0}^{t_1} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} & \text{for } n = 1. \end{cases} \end{aligned}$$

Using the Young's inequality, we get

$$\begin{aligned} \|\mathbf{e}^n\|_{\mathbb{D}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{D}}^2 + 2\Delta t \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}}^2 &\leq \Delta t \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}}^2 + C\Delta t^4 \int_{t_{n-1}}^{t_n} (\|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2) ds \\ &\quad + C\Delta t \|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{W}}^2 \|\hat{\mathbf{e}}^{n-1}\|_{\mathbb{D}}^2. \end{aligned}$$

Summing the above inequality from $n = 1$ to L for any $1 \leq L \leq N$ and using the regularity assumption the the weak solution, we have

$$\|\mathbf{e}^L\|_{\mathbb{D}}^2 + \Delta t \sum_{n=1}^L \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}}^2 \leq C\Delta t^4 \int_0^T (\|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2) ds + C\Delta t \sum_{n=1}^L \|\mathbf{e}^n\|_{\mathbb{D}}^2.$$

Then, the discrete Grönwall inequality implies Eq. (53).

Taking the test function $\mathcal{Y} = \delta_t \mathbf{e}^n \in \mathbb{V}$ in the first equation of Eq. (56), we obtain

$$\begin{aligned}
\|\mathbf{e}^n\|_{\mathbb{A}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{A}}^2 + 2\Delta t \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 &= 2\Delta t R_1(\delta_t \mathbf{e}^n) + 2\Delta t R_2(\delta_t \mathbf{e}^n) - 2\Delta t [\mathbb{B}(\mathcal{X}(\cdot, t_{n-1/2}), \mathcal{X}(\cdot, t_{n-1/2}), \delta_t \mathbf{e}^n) - \mathbb{B}(\hat{\mathcal{X}}^{n-1}, \bar{\mathcal{X}}^{n-1/2}, \delta_t \mathbf{e}^n)] \\
&\leq C\Delta t^{5/2} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 ds \right)^{1/2} + C\Delta t^{7/2} \|\delta_t \mathbf{e}^n\|_{\mathbb{A}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} \\
&\quad + C\Delta t^{5/2} \|\delta_t \mathbf{e}^n\|_{\mathbb{A}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} + C\Delta t^{5/2} \|\mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{W}} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} \left(\int_{t_{n-1}}^{t_n} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 ds \right)^{1/2} \\
&\quad + C\Delta t \|\mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{W}} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} + C\Delta t \|\hat{\mathbf{e}}^{n-1}\|_{\mathbb{A}} \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{W}} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} \\
&\quad + \begin{cases} C\Delta t^{5/2} \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{V}} \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} \left(\int_{t_{n-2}}^{t_n} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 ds \right)^{1/2} & \text{for } n \geq 2, \\ C\|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{V}} \|\delta_t \mathbf{e}^1\|_{\mathbb{D}} \left(\Delta t^7 \int_{t_0}^{t_1} \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 ds + \Delta t^5 \int_{t_0}^{t_1} \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 ds \right)^{1/2} & \text{for } n = 1. \end{cases}
\end{aligned}$$

The Young's inequality implies

$$\begin{aligned}
\|\mathbf{e}^n\|_{\mathbb{A}}^2 - \|\mathbf{e}^{n-1}\|_{\mathbb{A}}^2 + 2\Delta t \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 &\leq \Delta t \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 + \Delta t^4 \int_{t_{n-1}}^{t_n} (\|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2) ds \\
&\quad + C\Delta t^{3/2} (\|\mathbf{e}^n\|_{\mathbb{A}} + \|\mathbf{e}^{n-1}\|_{\mathbb{A}}) \left(\int_{t_{n-1}}^{t_n} (\|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2) ds \right)^{1/2} + C\Delta t \|\bar{\mathcal{X}}^{n-1/2}\|_{\mathbb{W}}^2 \|\bar{\mathbf{e}}^{n-1}\|_{\mathbb{A}}^2 \\
&\quad + C\Delta t \|\mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{W}}^2 (\|\mathbf{e}^n\|_{\mathbb{A}}^2 + \|\mathbf{e}^{n-1}\|_{\mathbb{A}}^2).
\end{aligned}$$

Summing the above equation from $n = 1$ to L for any $1 \leq L \leq N$, we have

$$\|\mathbf{e}^L\|_{\mathbb{A}}^2 + \Delta t \sum_{n=1}^L \|\delta_t \mathbf{e}^n\|_{\mathbb{D}}^2 \leq \Delta t^4 \int_0^T (\|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{D}}^2 + \|\partial_{tt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{A}}^2 + \|\partial_{ttt} \mathcal{X}(\cdot, s)\|_{\mathbb{W}}^2) ds + C\Delta t \sum_{n=1}^L \|\mathbf{e}^n\|_{\mathbb{A}}^2.$$

The discrete Grönwall inequality implies Eq. (54).

The inf-sup condition (28) and the first equation of (56) imply

$$\begin{aligned}
\|\bar{\mathbf{e}}_p^{n-1/2}\| &\lesssim \sup_{\mathcal{Y} \in \mathbb{V} \setminus \{0\}} \frac{\mathbf{c}(\mathcal{Y}, \bar{\mathbf{e}}_p^{n-1/2})}{\|\mathcal{Y}\|_{\mathbb{V}}} \\
&= \sup_{\mathcal{Y} \in \mathbb{V} \setminus \{0\}} \frac{-\mathbb{D}(\delta_t \mathbf{e}^n, \mathcal{Y}) - \mathbb{A}(\bar{\mathbf{e}}^{n-1/2}, \mathcal{Y}) + R_1(\mathcal{Y}) + R_2(\mathcal{Y}) + R_3(\mathcal{Y}) - [\mathbb{B}(\mathcal{X}(\cdot, t_{n-1/2}), \mathcal{X}(\cdot, t_{n-1/2}), \mathcal{Y}) - \mathbb{B}(\mathcal{X}^{n-1}, \bar{\mathcal{X}}^{n-1/2}, \mathcal{Y})]}{\|\mathcal{Y}\|_{\mathbb{V}}} \\
&\lesssim \|\delta_t \mathbf{e}^n\|_{\mathbb{D}} + \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}} + \|\delta_t \mathcal{X}(\cdot, t_{n-1/2}) - \mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{D}} + \|\bar{\mathcal{X}}(\cdot, t_{n-1/2}) - \mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{A}} + \|\bar{\bar{p}}(\cdot, t_{n-1/2}) - \bar{p}(\cdot, t_{n-1/2})\| \\
&\quad + \|\mathcal{X}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\mathcal{X}(\cdot, t_{n-1/2}) - \bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{V}} + \|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{V}} \|\mathcal{X}(\cdot, t_{n-1/2}) - \mathcal{X}(\cdot, t_{n-1})\|_{\mathbb{V}} \\
&\quad + \|\bar{\mathcal{X}}(\cdot, t_{n-1/2})\|_{\mathbb{W}} \|\mathbf{e}^{n-1}\|_{\mathbb{D}} + \|\mathcal{X}^{n-1}\|_{\mathbb{V}} \|\bar{\mathbf{e}}^{n-1/2}\|_{\mathbb{A}}.
\end{aligned}$$

Then, Eq. (55) follows by using (53) and (54). $\square$

5.2 Spatial discretization

For any $\mathcal{X}_h = (\mathbf{u}_h, \boldsymbol{\omega}_h, \mathbf{m}_h, \varphi_h)^\top \in \mathbb{V}_h^2$, $\mathcal{Y}_h = (\mathbf{v}_h, \mathbf{s}_h, \mathbf{F}_h, \psi_h)^\top \in \mathbb{V}_h^2$ and $\mathcal{Z}_h = (\boldsymbol{\epsilon}_h, \boldsymbol{\sigma}_h, \mathbf{A}_h, \phi_h)^\top \in \mathbb{V}_h^2$, we define the discrete bilinear form $\mathbf{A}_h^2 : \mathbb{V}_h^2 \times \mathbb{V}_h^2 \rightarrow \mathbb{R}$ as:

$$\begin{aligned}
\mathbf{A}_h^2(\mathcal{X}_h, \mathcal{Y}_h) &:= \eta(\nabla \mathbf{u}_h, \nabla \mathbf{v}_h) + \eta'(\nabla \boldsymbol{\omega}_h, \nabla \mathbf{s}_h) + (\eta' + \lambda')(\operatorname{div} \boldsymbol{\omega}_h, \operatorname{div} \mathbf{s}_h) + \sigma(\operatorname{curl}_h^2 \mathbf{m}_h, \operatorname{curl}_h^2 \mathbf{F}_h) + \sigma(\operatorname{div} \mathbf{m}_h, \operatorname{div} \mathbf{F}_h) \\
&\quad + \frac{1}{\tau}(\mathbf{m}_h, \mathbf{F}_h) + \frac{\mu_0(1 + \chi_0) + \chi_0}{\tau}(\nabla \varphi_h, \nabla \psi_h) + \sigma\mu_0(\Delta_h^2 \varphi_h, \Delta_h^2 \psi_h) + \zeta(\operatorname{curl} \mathbf{u}_h, \operatorname{curl} \mathbf{v}_h) - 2\zeta(\boldsymbol{\omega}_h, \operatorname{curl} \mathbf{v}_h) \\
&\quad - 2\zeta(\operatorname{curl} \mathbf{u}_h, \mathbf{s}_h) + 4\zeta(\boldsymbol{\omega}_h, \mathbf{s}_h),
\end{aligned}$$

and define the discrete trilinear form $\mathbf{B}_h^2 : \mathbb{V}_h^2 \times \mathbb{V}_h^2 \times \mathbb{V}_h^2 \rightarrow \mathbb{R}$ as:

$$\begin{aligned} \mathbf{B}_h^2(\mathcal{Z}_h, \mathcal{X}_h, \mathcal{Y}_h) := & \rho b(\epsilon_h, \mathbf{u}_h, \mathbf{v}_h) + \rho \kappa b(\epsilon_h, \boldsymbol{\omega}_h, \mathbf{s}_h) - c(\epsilon_h, \mathbf{m}_h, \mathbf{F}_h) - \frac{1}{2}(\mathbf{m}_h \times \text{curl } \epsilon_h, \mathbf{F}_h) - (\boldsymbol{\sigma}_h \times \mathbf{m}_h, \mathbf{F}_h) \\ & + \mu_0[(\nabla \varphi_h, \mathbf{v}_h \text{ div } \mathbf{A}_h) - (\nabla \psi_h, \mathbf{u}_h \text{ div } \mathbf{A}_h)] + \frac{\mu_0}{2}[\mathbf{u}_h \times \text{curl } \mathbf{A}_h, \nabla \psi_h) - (\mathbf{v}_h \times \text{curl } \mathbf{A}_h, \nabla \varphi_h)] \\ & + \frac{\mu_0}{2}[(\mathbf{A}_h \times \text{curl } \mathbf{u}_h, \nabla \psi_h) - (\mathbf{A}_h \times \text{curl } \mathbf{v}_h, \nabla \varphi_h)] + \mu_0[(\boldsymbol{\omega}_h \times \mathbf{A}_h, \nabla \psi_h) - (\mathbf{s}_h \times \mathbf{A}_h, \nabla \varphi_h)] \\ & + \frac{\chi_0}{\tau}[(\mathbf{m}_h, \nabla \psi_h) - (\mathbf{F}_h, \nabla \varphi_h)] + \frac{1}{2}[(\epsilon_h \times \mathbf{F}_h, \text{curl}_h^2 \mathbf{m}_h) - (\epsilon_h \times \mathbf{m}_h, \text{curl}_h^2 \mathbf{F}_h)] \\ & + \frac{\mu_0}{2}[(\Delta_h^2 \psi_h, \mathbf{u}_h \cdot \mathbf{A}_h) - (\Delta_h^2 \varphi_h, \mathbf{v}_h \cdot \mathbf{A}_h)]. \end{aligned}$$

Obviously, $\mathbf{B}_h^2$ is skew-symmetric with respect to the second and third variables, i.e.,

$$\mathbf{B}_h^2(\mathcal{Z}_h, \mathcal{X}_h, \mathcal{Y}_h) = -\mathbf{B}_h^2(\mathcal{Z}_h, \mathcal{Y}_h, \mathcal{X}_h) \quad \forall \mathcal{X}_h, \mathcal{Z}_h, \mathcal{Y}_h \in \mathbb{V}_h^2. \quad (57)$$

For any $1 \leq n \leq N$, the second order abstract formulation of the fully discrete schemes of (2) read as:

Algorithm 5 Second Order Scheme (fully discretization)

Given $\mathcal{X}_h^{n-1} \in \mathbb{V}_h^2$, find $\mathcal{X}_h^n \in \mathbb{V}_h^2$ and $\tilde{p}_h^n \in L_h$, such that

$$\begin{cases} \mathbf{D}(\delta_t \mathcal{X}_h^n, \mathcal{Y}_h) + \mathbf{A}_h^2(\bar{\mathcal{X}}_h^{n-1/2}, \mathcal{Y}_h) + \mathbf{B}_h^2(\hat{\mathcal{X}}_h^{n-1}, \bar{\mathcal{X}}_h^{n-1/2}, \mathcal{Y}_h) + \mathbf{C}(\mathcal{Y}_h, \tilde{p}_h^{n-1/2}) = (\mathbf{f}(\cdot, t_{n-1/2}), \mathcal{Y}_h) & \forall \mathcal{Y}_h \in \mathbb{V}_h^2, \\ \mathbf{C}(\bar{\mathcal{X}}_h^{n-1/2}, q_h) = 0 & \forall q_h \in L_h, \end{cases} \quad (58)$$

with initial value

$$\mathcal{X}_h^0(\cdot) = \mathcal{I}_h^2 \mathcal{X}_0(\cdot).$$

Similar to the analysis of Algorithms 1 and 4, we obtain the following existence and uniqueness result for the fully discrete scheme Algorithm 5.

Theorem 5.4. *For any $2 \leq n \leq N$, given $\mathcal{X}_h^{n-1} \in \mathbb{V}_h^2$ and $\mathcal{X}_h^{n-2} \in \mathbb{V}_h^2$, the fully discrete scheme Algorithm 5 admits a unique solution $(\mathcal{X}_h^n, \tilde{p}_h^n) \in \mathbb{V}_h^2 \times L_h$ at each time step.*

Theorem 5.5 (Energy estimate for Algorithm 5). *Given $\mathcal{X}_h^{n-1} \in \mathbb{V}_h^2$ for $1 \leq n \leq N$, let $(\mathcal{X}_h^n, \tilde{p}_h^n) \in \mathbb{V}_h^2 \times L_h$ be the unique solution of Algorithm 5. Then the following energy estimate holds:*

$$\left(\max_{0 \leq n \leq N} \|\mathcal{X}_h^n\|_{\mathbb{D}}^2 + 2\Delta t \sum_{n=1}^N \|\bar{\mathcal{X}}_h^{n-1/2}\|_{\mathbb{A}}^2 \right)^{1/2} \leq \|\mathcal{X}_h^0\|_{\mathbb{D}} + 2\Delta t \sum_{n=1}^N \|\mathbf{f}(\cdot, t_{n-1/2})\|_{\mathbb{D}^{-1}}. \quad (59)$$

At each time level t_n , for any discrete quantity $\mathbf{v}_h$, we use the notation

$$\bar{\mathbf{v}}_h := \bar{\mathbf{v}}_h^{n-1/2}, \quad \hat{\mathbf{v}}_h^- := \hat{\mathbf{v}}_h^{n-1}, \quad \mathbf{v}_h := \mathbf{v}_h^n, \quad \delta_t \mathbf{v}_h := \delta_t \mathbf{v}_h^n.$$

With this convention, the fully discrete scheme (58) in Algorithm 5 can be equivalently rewritten in the following computable form. Find

$$\begin{aligned} \mathbf{u}_h \in \mathbf{S}_h^2, \quad \tilde{p}_h \in L_h, \quad \boldsymbol{\omega}_h \in \boldsymbol{\Sigma}_h^2, \quad \mathbf{m}_h \in \mathbf{V}_h^2, \\ \mathbf{z}_h \in \mathbf{U}_h^2, \quad \mathbf{k}_h \in \mathbf{U}_h^2, \quad \varphi_h \in \Sigma_h^2, \quad \vartheta_h \in \Sigma_h^2, \quad \varpi_h \in \Sigma_h^2, \end{aligned}$$

such that, for all

$$\begin{aligned} \mathbf{v}_h \in \mathbf{S}_h^2, \quad \tilde{q}_h \in L_h, \quad \mathbf{s}_h \in \boldsymbol{\Sigma}_h^2, \quad \mathbf{F}_h \in \mathbf{V}_h^2, \\ \boldsymbol{\lambda}_h \in \mathbf{U}_h^2, \quad \boldsymbol{\theta}_h \in \mathbf{U}_h^2, \quad \psi_h \in \Sigma_h^2, \quad \varsigma_h \in \Sigma_h^2, \quad \gamma_h \in \Sigma_h^2, \end{aligned}$$

it holds

$$\left\{ \begin{array}{l} \rho(\delta_t \mathbf{u}_h, \mathbf{v}_h) + \rho b(\hat{\mathbf{u}}_h^-, \bar{\mathbf{u}}_h, \mathbf{v}_h) + \eta(\nabla \bar{\mathbf{u}}_h, \nabla \mathbf{v}_h) + \zeta(\text{curl } \bar{\mathbf{u}}_h, \text{curl } \mathbf{v}_h) - 2\zeta(\bar{\boldsymbol{\omega}}_h, \text{curl } \mathbf{v}_h) - (\bar{p}_h, \text{div } \mathbf{v}_h) \\ \quad + \frac{\mu_0}{2}(\bar{\vartheta}_h, \hat{\mathbf{m}}_h^- \cdot \mathbf{v}_h) + \frac{\mu_0}{2}(\nabla \bar{\varphi}_h \text{div } \hat{\mathbf{m}}_h^-, \mathbf{v}_h) - \frac{\mu_0}{2}(\mathbf{v}_h \times \hat{\mathbf{k}}_h^-, \nabla \bar{\varphi}_h) - \frac{\mu_0}{2}(\hat{\mathbf{m}}_h^- \times \text{curl } \mathbf{v}_h, \nabla \bar{\varphi}_h) = 0, \\ \quad (\text{div } \bar{\mathbf{u}}_h, q_h) = 0, \\ \rho\kappa(\delta_t \bar{\boldsymbol{\omega}}_h, \mathbf{s}_h) + \eta'(\nabla \bar{\boldsymbol{\omega}}_h, \nabla \mathbf{s}_h) + (\eta' + \lambda')(\text{div } \bar{\boldsymbol{\omega}}_h, \text{div } \mathbf{s}_h) + 4\zeta(\bar{\boldsymbol{\omega}}_h, \mathbf{s}_h) + \rho\kappa b(\hat{\mathbf{u}}_h^-, \bar{\boldsymbol{\omega}}_h, \mathbf{s}_h) \\ \quad - \mu_0(\hat{\mathbf{m}}_h^- \times \nabla \bar{\varphi}_h, \mathbf{s}_h) - 2\zeta(\text{curl } \bar{\mathbf{u}}_h, \mathbf{s}_h) = 0, \\ (\delta_t \mathbf{m}_h, \mathbf{F}_h) + \frac{1}{\tau}(\bar{\mathbf{m}}_h, \mathbf{F}_h) + \sigma(\text{curl } \bar{\mathbf{k}}_h, \mathbf{F}_h) + \sigma(\text{div } \bar{\mathbf{m}}_h, \text{div } \mathbf{F}_h) - c(\hat{\mathbf{u}}_h^-, \bar{\mathbf{m}}_h, \mathbf{F}_h) - \frac{1}{2}(\bar{\mathbf{m}}_h \times \text{curl } \hat{\mathbf{u}}_h^-, \mathbf{F}_h) \\ \quad - \frac{1}{2}(\hat{\mathbf{u}}_h^- \times \bar{\mathbf{k}}_h, \mathbf{F}_h) - \frac{1}{2}(\text{curl } \bar{\mathbf{z}}_h, \mathbf{F}_h) - (\bar{\boldsymbol{\omega}}_h^- \times \bar{\mathbf{m}}_h, \mathbf{F}_h) - \frac{\chi_0}{\tau}(\nabla \bar{\varphi}_h, \mathbf{F}_h) = 0, \\ \quad (\mathbf{z}_h, \boldsymbol{\lambda}_h) - (\hat{\mathbf{u}}_h^- \times \bar{\mathbf{m}}_h, \boldsymbol{\lambda}_h) = 0, \\ \quad (\bar{\mathbf{k}}_h, \boldsymbol{\theta}_h) - (\bar{\mathbf{m}}_h, \text{curl } \boldsymbol{\theta}_h) = 0, \\ \mu_0(\delta_t \nabla \bar{\varphi}_h, \nabla \psi_h) + \frac{\mu_0(1+\chi_0)+\chi_0}{\tau}(\nabla \bar{\varphi}_h, \nabla \psi_h) + \sigma\mu_0(\nabla \bar{\vartheta}_h, \nabla \psi_h) + \mu_0(\bar{\boldsymbol{\omega}}_h \times \hat{\mathbf{m}}_h^-, \nabla \psi_h) + \frac{\chi_0}{\tau}(\bar{\mathbf{m}}_h, \nabla \psi_h) \\ \quad - \frac{\mu_0}{2}(\nabla \bar{\boldsymbol{\omega}}_h, \nabla \psi_h) - \frac{\mu_0}{2}(\bar{\mathbf{u}}_h \text{div } \hat{\mathbf{m}}_h^-, \nabla \psi_h) + \frac{\mu_0}{2}(\hat{\mathbf{m}}_h^- \times \text{curl } \bar{\mathbf{u}}_h, \nabla \psi_h) + \frac{\mu_0}{2}(\bar{\mathbf{u}}_h \times \hat{\mathbf{k}}_h^-, \nabla \psi_h) \\ \quad + (\partial_t \mathbf{H}_e(t_{n-1/2}), \nabla \psi_h) + \frac{\mu_0+\chi_0}{\tau\mu_0}(\mathbf{H}_e(t_n - 1/2), \nabla \psi_h) + \sigma(\text{grad div } \mathbf{H}_e(t_n - 1/2), \nabla \psi_h) = 0, \\ \quad \sigma\mu_0(\bar{\vartheta}_h, \varsigma_h) - \sigma\mu_0(\nabla \bar{\varphi}_h, \nabla \varsigma_h) = 0, \\ \quad (\bar{\boldsymbol{\omega}}_h, \boldsymbol{\gamma}_h) - (\bar{\mathbf{u}}_h \cdot \hat{\mathbf{m}}_h^-, \boldsymbol{\gamma}_h) = 0. \end{array} \right. \quad (60)$$

After solving (60), we can obtain $\mathbf{H}_h \in \mathbf{U}_h^2$ by solving

$$(\mathbf{H}_h, \boldsymbol{\varepsilon}_h) = (\nabla \varphi_h, \boldsymbol{\varepsilon}_h) \quad \forall \boldsymbol{\varepsilon}_h \in \mathbf{U}_h^2. \quad (61)$$

Consequently, $\mathbf{H}_h = \nabla \varphi_h$ and therefore $\text{curl } \mathbf{H}_h = 0$ pointwise in Ω .

Remark 4. When $n = 1$, the fully discrete system (60) is nonlinear. To solve the resulting nonlinear algebraic system, we employ the quasi-Newton iteration developed in [32, 34]. For $2 \leq n \leq N$, owing to the extrapolation treatment of the nonlinear terms, the system (60) becomes linear and has the same block algebraic structure as the first-order scheme (47). Consequently, the linear systems arising at these time levels can be solved efficiently by the exact Schur complement decoupling solver presented in Algorithm 3.

The fully discrete error analysis of the second-order scheme follows essentially the same strategy as that of the first-order method. By introducing suitable quasi-interpolation operators, the spatial approximation errors can be treated in a standard manner, while the main additional issue is the consistency error associated with the discrete differential operators. Since no essentially new arguments are required, we omit the technical details here for brevity and refer to the arXiv version of this paper for the complete fully discrete error analysis.

6 Numerical experiments

We first collect the convergence and unforced energy-dissipation experiments for both time discretizations, and then present two magnetically forced channel-flow applications of the second-order scheme.

6.1 Experimental setup

We assess both time discretizations by a manufactured-solution test on $\Omega = (0, 1)^3$ with $T = 2$. Each Cartesian partition into $K \times K \times K$ cubes is subdivided into tetrahedra, with $K = 4, 8, 16, 32$ and $\Delta t = 1/K = h/\sqrt{3}$, where h is the maximum tetrahedron diameter. Space and time are refined together; this experiment does not isolate the temporal convergence order.

Manufactured solution. Define

$$q(s) = 64s^3(1-s)^3, \quad B(x, y, z) = q(x)q(y)q(z), \quad S(x, y, z) = \sin^2(\pi x) \sin^2(\pi y) \sin^2(\pi z),$$

and let $\mathbf{a} = (1, 2, 3)^\top$, $\mathbf{b} = (1, -1, 1/2)^\top$, and $\mathbf{c} = (1, 2, -1)^\top$. The prescribed fields are

$$\begin{aligned} \mathbf{u} &= \frac{1+0.1t}{C_u} \nabla B \times \mathbf{a}, & \boldsymbol{\omega} &= \frac{1-0.05t}{C_\omega} B \mathbf{b}, & \mathbf{m} &= \frac{1+0.05t}{C_m} S \mathbf{c}, \\ \varphi &= \frac{1-0.05t}{C_\varphi} B, & \mathbf{H} &= \nabla \varphi, & p &= \frac{1+0.05t}{C_p} \prod_{s \in \{x, y, z\}} \cos(2\pi s), \end{aligned}$$

where the fixed normalization constants used in the computations are

$$\begin{aligned} C_u &= 363.275188821, & C_\omega &= 20.6500992827, & C_m &= \frac{\sqrt{81+108\pi^2}}{16}, \\ C_\varphi &= 13.7667328551, & C_p &= 0.353553390593. \end{aligned}$$

The auxiliary fields are $\mathbf{k} = \text{curl } \mathbf{m}$, $\mathbf{z} = \mathbf{u} \times \mathbf{m}$, $\vartheta = -\Delta\varphi$, and $\varpi = \mathbf{u} \cdot \mathbf{m}$. We set $\mathbf{H}_e = -\mu_0(\mathbf{m} + \mathbf{H})$ and use the zero-mean modified pressure $\tilde{p} = \mathcal{P}_0[p - \mu_0 \mathbf{m} \cdot \mathbf{H}/2]$, where $\mathcal{P}_0 v = v - |\Omega|^{-1} \int_{\Omega} v$. All physical parameters $\rho, \kappa, \eta, \zeta, \mu_0, \sigma, \eta', \lambda', \tau, \chi_0$ are set to one.

This is a forced verification problem. The momentum, angular-momentum, and magnetization sources $\mathbf{f}, \mathbf{g}, \mathbf{r}$ are obtained by substituting the prescribed fields into the corresponding continuous equations. Their weak right-hand sides are $(\mathbf{f}, \mathbf{v}_h)$, $(\mathbf{g}, \mathbf{s}_h)$, and $(\mathbf{r}, \mathbf{F}_h)$; the magnetic-potential right-hand side is

$$\left(-\partial_t \mathbf{H}_e - \frac{\mu_0 + \chi_0}{\tau \mu_0} \mathbf{H}_e + \sigma \nabla \text{div } \mathbf{H}_e - \mu_0 \mathbf{r}, \nabla \psi_h \right).$$

The fields satisfy the homogeneous boundary traces used in the experiment: zero velocity and angular velocity, zero normal magnetization, zero tangential $\mathbf{z}, \mathbf{k}, \mathbf{H}$, and zero $\varphi, \vartheta, \varpi$. In particular, q, q', q'' vanish at both endpoints and $S, \nabla S$ vanish on the cube boundary. The new ϑ and ϖ couplings are not identically zero for this test.

Discretization and implementation. The first-order runs use the MINI velocity element, continuous piecewise linear pressure and angular velocity, lowest-order Raviart–Thomas magnetization and first-family Nedelec $\mathbf{z}, \mathbf{k}, \mathbf{H}$, and continuous piecewise linear $\varphi, \vartheta, \varpi$.

For the second-order scheme, velocity and angular velocity use continuous piecewise quadratic elements, pressure remains continuous piecewise linear, magnetization uses RT1, and $\mathbf{z}, \mathbf{k}, \mathbf{H}$ use first-family Nedelec elements one degree above the lowest order. The scalar fields $\varphi, \vartheta, \varpi$ are continuous piecewise quadratic. The corresponding RT and Nedelec Basix degree is two.

The initial velocity is obtained by a discrete Stokes projection. After interpolating the initial magnetization into its RT space, the initial potential is determined from the compatible magnetic Poisson problem. The initial $\mathbf{k}, \mathbf{z}, \vartheta, \varpi$ satisfy their discrete mass-projection relations. Pressure is normalized by its finite-element integral mean.

For the second-order convergence test, the first time step is solved by a Broyden quasi-Newton iteration with relative tolerance 10^{-9} and a maximum of 12 iterations. The accepted state is checked by reassembling the nonlinear midpoint equations. Subsequent time steps use the prescribed extrapolated coefficients and require one five-block linear solve per step. The $\mathbf{z}_h$ and ϖ_h slots represent the current midpoint-stage projections; they are not averaged again with the preceding stage.

Computations use FEniCSx/DOLFINx 0.10.0 [8] with PETSc and MUMPS. The five-block Algorithm 3 is evaluated through matrix-free Schur-complement actions. The K_1, K_3, K_5 subproblems use MUMPS LU; the S_2, S_4 systems use FGMRES with LU preconditioners. The MUMPS ordering control is ICNTL(7) = 7. The Schur tolerances are 10^{-10} relative and 10^{-12} absolute. All first-order cases and the second-order cases up to $K = 16$ use 48 MPI processes with one thread each; the second-order $K = 32$ case uses 32 MPI processes with four threads each on a 1-TB-class node. Source terms are evaluated at quadrature points, with assembly quadrature degree 12. In the error calculations, the analytic values and derivatives are independently represented in a degree-6 DG space and integrated with degree-12 quadrature; the numerical solution is evaluated in its original finite-element space.

As implementation checks for the manufactured-solution runs, the two schemes were compared with a monolithic solve of the same matrix at the first two time levels on $K = 4$, modulo the same pressure nullspace; the normalized solution differences were below 1.1×10^{-12} . An independent transcription of the nine weak equations on $K = 2$, using random finite-element fields and both unit and non-unit parameters, also agreed with the production matrices and right-hand sides to scaled differences below 1.1×10^{-16} . These checks concern algebraic and implementation consistency, rather than replacing the convergence test.

Error measures. For a field v and its specified norm X , we report

$$E_X(v) = \frac{\|v - v_h\|_X}{\|v\|_X}, \quad \text{rate}(K \rightarrow 2K) = \log_2 \frac{E_X(v; K)}{E_X(v; 2K)}.$$

The norms are full H^1 for $\mathbf{u}, \boldsymbol{\omega}, \varphi$, L^2 for $\tilde{p}, \mathbf{k}$, $H(\text{div})$ for $\mathbf{m}$, and $H(\text{curl})$ for $\mathbf{H}$. Thus the graph norms combine the squared L^2 and derivative terms before taking the square root. Pressure errors use zero-mean numerical and exact fields. First-order errors are evaluated at T .

For the second-order scheme, the primary fields are compared at T . Its pressure slot represents the averaged modified pressure, so its reference is

$$\tilde{p}_{\text{ref}}^{N-1/2} = \frac{\tilde{p}(T) + \tilde{p}(T - \Delta t)}{2},$$

with the same zero-mean convention as above. This reference is not identified with either the endpoint pressure or the analytic pressure evaluated directly at the midpoint.

We also report C_H , the maximum of $|\text{curl } \mathbf{H}_h|$ over the degree-6 DG interpolation points, as a sampled estimate of $\|\text{curl } \mathbf{H}_h\|_{L^\infty}$. It is an absolute diagnostic, not a relative error or a rigorous continuous supremum.

6.2 Convergence verification

We report the joint space–time refinement results separately for the two schemes, using the common test and error measures defined above.

6.2.1 First-order scheme

Table 1 gives the errors and observed orders; Figure 1 shows the corresponding log-log error curves. On the finest refinement interval, the rates for $\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \mathbf{k}, \varphi, \mathbf{H}$ are 1.019, 0.994, 0.999, 0.997, 1.018, and 1.017, respectively. The modified pressure exhibits a rate of 2.065 in this test. The sampled curl diagnostic remains between 1.22×10^{-13} and 8.51×10^{-12} , consistent with the discrete gradient recovery of the magnetic field up to numerical error.

Table 1: Relative errors and observed rates for the first-order scheme, $T = 2$ and $\Delta t = 1/K$. C_H is the sampled absolute curl diagnostic.

(a) Velocity, pressure, angular velocity, and magnetization								
K	$\mathbf{u}: H^1$		$\tilde{p}: L^2$		$\boldsymbol{\omega}: H^1$		$\mathbf{m}: H(\text{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
4	9.6554×10^{-1}	–	5.6019×10^{-1}	–	6.4182×10^{-1}	–	3.9160×10^{-1}	–
8	4.3070×10^{-1}	1.165	1.0260×10^{-1}	2.449	3.5671×10^{-1}	0.847	1.9873×10^{-1}	0.979
16	1.9858×10^{-1}	1.117	2.1172×10^{-2}	2.277	1.8331×10^{-1}	0.960	9.9738×10^{-2}	0.995
32	9.8002×10^{-2}	1.019	5.0596×10^{-3}	2.065	9.2018×10^{-2}	0.994	4.9919×10^{-2}	0.999

(b) Auxiliary curl field, potential, magnetic field, and curl diagnostic								
K	$\mathbf{k}: L^2$		$\varphi: H^1$		$\mathbf{H}: H(\text{curl})$		C_H	
	Error	Rate	Error	Rate	Error	Rate		
4	5.1511×10^{-1}	–	6.5848×10^{-1}	–	6.6223×10^{-1}	–	1.2220×10^{-13}	
8	2.6813×10^{-1}	0.942	3.7678×10^{-1}	0.805	3.7965×10^{-1}	0.803	5.1946×10^{-13}	
16	1.3523×10^{-1}	0.988	1.8757×10^{-1}	1.006	1.8935×10^{-1}	1.004	2.0267×10^{-12}	
32	6.7750×10^{-2}	0.997	9.2620×10^{-2}	1.018	9.3572×10^{-2}	1.017	8.5067×10^{-12}	

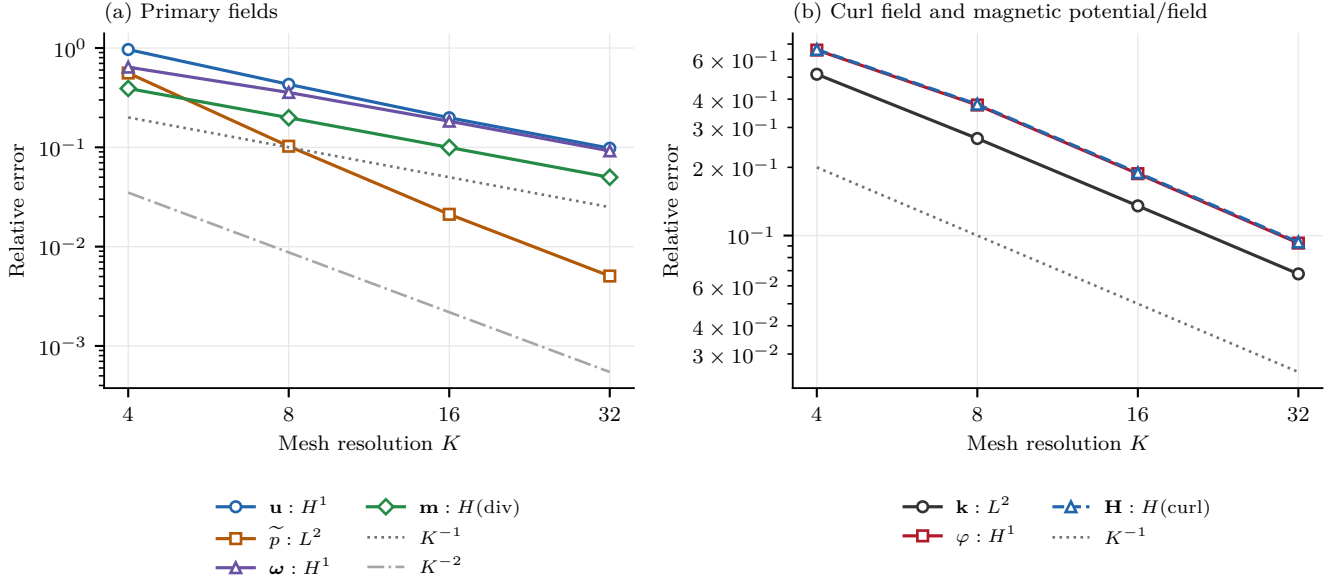


Figure 1: Relative errors for the first-order discretization under joint refinement with $\Delta t = 1/K$. (a) Velocity, modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is the mesh resolution $K = 4, 8, 16, 32$, so refinement proceeds from left to right. Reference lines proportional to K^{-1} and K^{-2} indicate first- and second-order decay with mesh refinement, not independently measured temporal orders.

6.2.2 Second-order scheme

Table 2 and Fig. 2 show that the finest-grid rates for $\mathbf{u}, \tilde{p}, \boldsymbol{\omega}, \mathbf{m}, \varphi, \mathbf{H}$ are 2.166, 2.051, 1.981, 2.007, 1.968, and 1.968, respectively. The L^2 error of $\mathbf{k}$ decreases approximately at first order, with a finest-grid rate of 1.015; second-order convergence is therefore not claimed for every auxiliary quantity. The sampled curl diagnostic ranges from 7.30×10^{-13} to 6.96×10^{-11} . For $K = 32$, all 64 steps reach $T = 2$, and the final full-system relative algebraic residual is 2.08×10^{-13} . These observations support the reported joint-refinement behavior. Because the mesh and time step vary together and the primary exact fields have linear temporal amplitudes, they do not constitute a separate test of pure temporal accuracy or of energy stability.

Table 2: Relative errors and observed rates for the second-order scheme, $T = 2$ and $\Delta t = 1/K$. C_H is the sampled absolute curl diagnostic.

(a) Velocity, pressure, angular velocity, and magnetization

K	$\mathbf{u}: H^1$		$\tilde{p}: L^2$		$\boldsymbol{\omega}: H^1$		$\mathbf{m}: H(\text{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
4	5.6873×10^{-1}	—	4.2000×10^{-1}	—	2.1381×10^{-1}	—	1.1937×10^{-1}	—
8	1.5090×10^{-1}	1.914	9.4815×10^{-2}	2.147	6.2017×10^{-2}	1.786	3.1613×10^{-2}	1.917
16	2.4805×10^{-2}	2.605	2.0872×10^{-2}	2.184	1.6320×10^{-2}	1.926	7.8383×10^{-3}	2.012
32	5.5259×10^{-3}	2.166	5.0382×10^{-3}	2.051	4.1345×10^{-3}	1.981	1.9494×10^{-3}	2.007

(b) Auxiliary curl field, potential, magnetic field, and curl diagnostic

K	$\mathbf{k}: L^2$		$\varphi: H^1$		$\mathbf{H}: H(\text{curl})$		C_H
	Error	Rate	Error	Rate	Error	Rate	
4	2.1008×10^{-1}	—	2.0119×10^{-1}	—	2.0303×10^{-1}	—	7.3001×10^{-13}
8	9.5804×10^{-2}	1.133	6.0098×10^{-2}	1.743	6.0718×10^{-2}	1.742	3.9064×10^{-12}
16	4.6084×10^{-2}	1.056	1.6119×10^{-2}	1.899	1.6289×10^{-2}	1.898	1.8492×10^{-11}
32	2.2804×10^{-2}	1.015	4.1189×10^{-3}	1.968	4.1627×10^{-3}	1.968	6.9611×10^{-11}

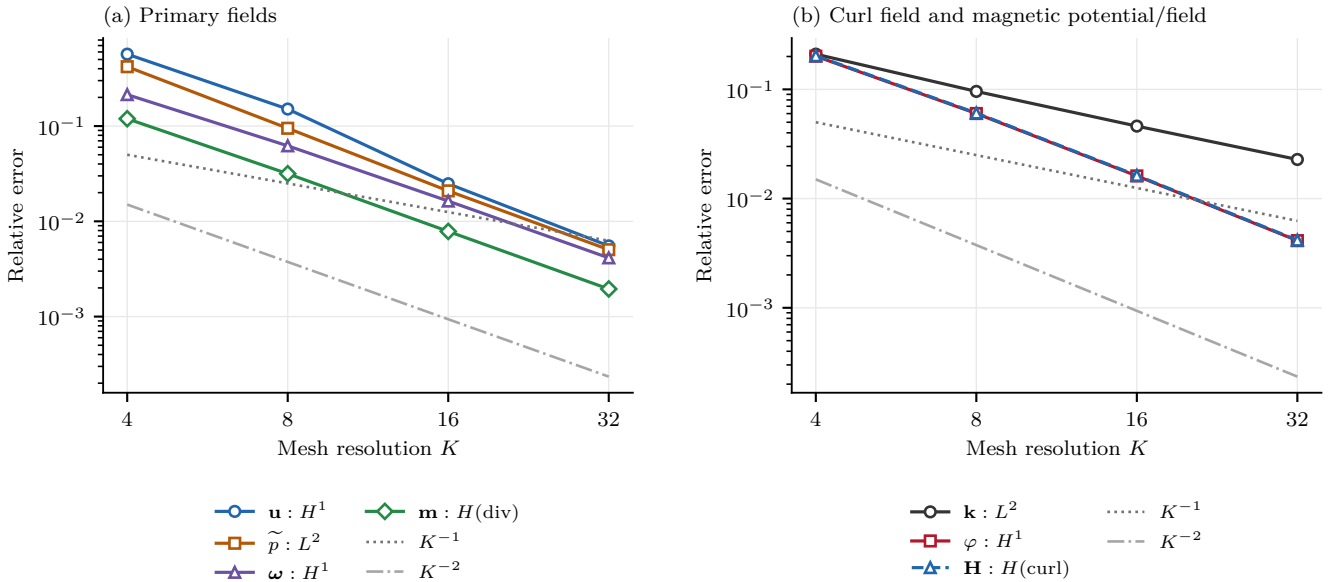


Figure 2: Relative errors for the second-order discretization using the same manufactured solution and $\Delta t = 1/K$. (a) Velocity, averaged modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is $K = 4, 8, 16, 32$, increasing from left to right. Reference lines proportional to K^{-1} and K^{-2} distinguish the nearly quadratic decay of the principal errors from the approximately first-order L^2 decay for $\mathbf{k}$.

6.3 Energy dissipation

We test free relaxation on the fixed $K = 8$ mesh, with $T = 0.5$ and $\Delta t = 1/16, 1/32, 1/64$. All physical parameters remain one. Unlike the manufactured-solution test, the applied field and all compensating sources are set to zero: $\mathbf{H}_e = \mathbf{f} = \mathbf{g} = \mathbf{r} = 0$. We retain the homogeneous boundary traces specified in Section 6.1. The initial shapes are

$$\mathbf{u}_0 = C_u^{-1} \nabla B \times \mathbf{a}, \quad \boldsymbol{\omega}_0 = C_\omega^{-1} B \mathbf{b}, \quad \mathbf{m}_0 = C_m^{-1} S \mathbf{c}.$$

The initial velocity is projected onto the discrete solenoidal space; angular velocity and magnetization are interpolated into their respective spaces. The initial potential is recomputed with zero boundary trace from

$$(\nabla \varphi_h^0, \nabla \psi_h) = -(\mathbf{m}_h^0, \nabla \psi_h) \quad \text{for all admissible } \psi_h,$$

rather than prescribed from the manufactured magnetic potential. The auxiliary variables satisfy their discrete projection relations. Thus the internal magnetic field is not artificially suppressed, and the subsequent solution is not constrained to follow the manufactured temporal profile.

At every time level we integrate the discrete energy

$$E_h^n = \frac{1}{2} \left(\rho \|\mathbf{u}_h^n\|^2 + \rho \kappa \|\boldsymbol{\omega}_h^n\|^2 + \|\mathbf{m}_h^n\|^2 + \mu_0 \|\nabla \varphi_h^n\|^2 \right). \quad (62)$$

The norms in (62) are spatial L^2 norms. Energy and dissipation are evaluated in the original finite-element spaces using degree-12 quadrature, not from the visualization fields. For the first-order scheme, we check the balance defect

$$R_h^n = E_h^n - E_h^{n-1} + \Delta t D_h^n + N_h^n, \quad N_h^n = \frac{1}{2} \|\mathcal{X}_h^n - \mathcal{X}_h^{n-1}\|_{\mathbb{D}}^2, \quad (63)$$

where D_h^n is the discrete viscous, spin, magnetic-diffusion, and relaxation dissipation corresponding to $\mathbf{A}$. In particular, the weak magnetic diffusion terms are evaluated as $\sigma \|\mathbf{k}_h^n\|^2$ and $\sigma \mu_0 \|\vartheta_h^n\|^2$. For the second-order discretization, the balance defect is

$$R_h^n = E_h^n - E_h^{n-1} + \Delta t D_h^{n-1/2}, \quad (64)$$

where $D_h^{n-1/2}$ is evaluated at the midpoint averages of the primary fields and the endpoint auxiliaries $\mathbf{k}_h, \vartheta_h$. There is no first-order numerical-dissipation term N_h^n . The stage fields $\mathbf{z}_h, \varpi_h$ are not averaged a second time.

For both schemes, the reported balance diagnostic is

$$\epsilon_{\text{bal}}^n = \frac{|R_h^n|}{\max\{E_h^n, E_h^{n-1}, \Delta t D_h^n + N_h^n, 10^{-14} E_h^0\}},$$

with the midpoint dissipation and $N_h^n = 0$ used for the second-order scheme.

Both schemes retain the spatial approximation orders specified in Section 6.1. Thus their discrete initial states differ, whereas each scheme uses the same initial state for all three time steps.

All energy runs use 48 MPI processes with one thread each and the same five-block solver settings as the convergence experiments. The first-order runs require no nonlinear startup.

For the second-order energy test, the initial Broyden solve uses relative tolerance 10^{-10} with at most 20 iterations, and the accepted midpoint equations are independently reassembled to check their residual. Subsequent time steps use the prescribed extrapolation.

6.3.1 First-order scheme

Figure 3 shows monotone decay for all three time steps. The initial discrete energy is approximately 4.205865×10^{-2} ; the final energy ratios and balance diagnostics are given in Table 3. The largest balance diagnostic is 4.61×10^{-14} . These results verify dissipation for the tested discretizations; they do not constitute a pure temporal convergence study.

Table 3: First-order unforced energy test: $K = 8$, $T = 0.5$.

Δt	$E_h(T)/E_h(0)$	$\max_n \epsilon_{\text{bal}}^n$
1/16	9.1847×10^{-5}	3.69×10^{-14}
1/32	3.1955×10^{-5}	2.22×10^{-14}
1/64	1.6247×10^{-5}	4.61×10^{-14}

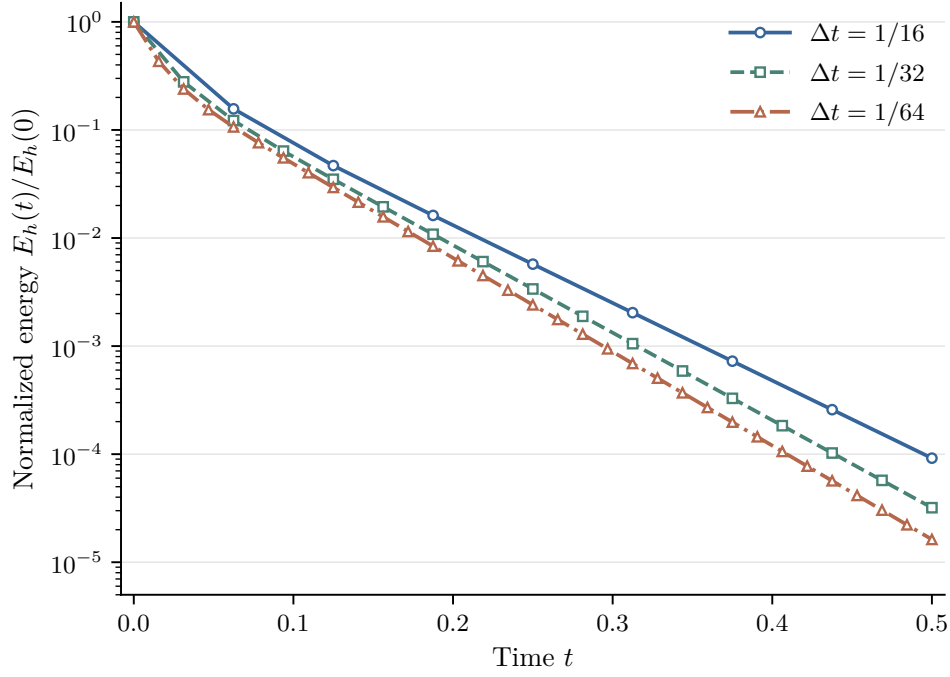


Figure 3: Normalized discrete energy for the first-order scheme in the unforced relaxation test, $K = 8$ and $T = 0.5$. The vertical axis is logarithmic. Symbols represent all computed time levels; connecting lines are not fitted or smoothed curves. The same discrete initial state is used for all three time steps.

6.3.2 Second-order scheme

Figure 4 shows monotone energy decay for all three time steps, with initial energy approximately 4.295033×10^{-2} . Table 4 reports the final ratios and balance diagnostics; the largest normalized balance defect is 3.71×10^{-11} . The coarsest time step produces slower late-time decay, which is retained in the plot. Energy dissipation therefore should not be interpreted as evidence that the temporal error is negligible. Since the two schemes also use different spatial approximation orders, these figures are separate stability tests rather than a controlled comparison of temporal accuracy. The computations provide numerical evidence of the discrete dissipation mechanism at the tested step sizes, not a proof for arbitrary step sizes.

Table 4: Second-order unforced energy test: $K = 8$, $T = 0.5$.

Δt	$E_h(T)/E_h(0)$	$\max_n \epsilon_{\text{bal}}^n$
1/16	1.1117×10^{-4}	3.71×10^{-11}
1/32	2.5984×10^{-5}	7.77×10^{-13}
1/64	1.0438×10^{-5}	2.67×10^{-13}

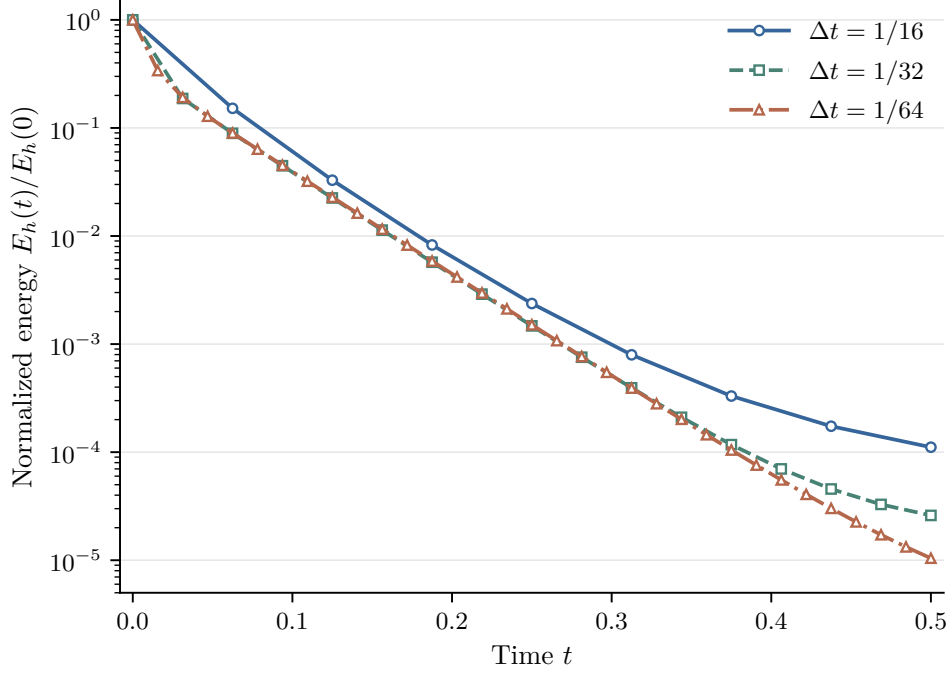


Figure 4: Normalized discrete energy for the second-order scheme in the unforced relaxation test, $K = 8$ and $T = 0.5$. The logarithmic vertical scale matches Figure 3. Symbols represent all computed time levels without smoothing. The same discrete initial state is used for all three time steps.

6.4 Magnetically forced flow in a straight channel

We next apply the second-order scheme and the five-block Algorithm 3 to three-dimensional ferrofluid transport with prescribed inflow. The configuration is motivated by the channel-pumping examples of Dong et al. [11], while retaining the magnetic diffusion, compatible finite element spaces, and midpoint/extrapolation scheme of the present work. The parameters and driving waveform below define our computational example; they are not a reproduction of all the parameters in that reference. All quantities are dimensionless.

Domain, material parameters, and initial data. The straight channel is

$$\Omega_c = (0, 1) \times (0, 0.2) \times (0, 0.2).$$

We denote the inlet $x = 0$, outlet $x = 1$, and remaining solid walls by Γ_i , Γ_o , and Γ_w , respectively. The material parameters, also used in the stepped channel below, are

$$\begin{aligned} \rho = \mu_0 = \chi_0 = 1, \quad \kappa = \zeta = 0.01, \quad \eta = 0.02, \\ \eta' = \lambda' = 0.001, \quad \tau = 0.1, \quad \sigma = 0.005. \end{aligned} \quad (65)$$

The body sources are $\mathbf{f} = \mathbf{g} = \mathbf{r} = \mathbf{0}$. The initial velocity, angular velocity, magnetization, and magnetic potential are zero, as are the pressure and auxiliary variables. These data are compatible with the initially vanishing inlet and magnetic forcing.

Inlet and magnetic forcing. Introduce the smooth ramp

$$R(s) = \begin{cases} 0, & s \leq 0, \\ 10s^3 - 15s^4 + 6s^5, & 0 < s < 1, \\ 1, & s \geq 1. \end{cases} \quad (66)$$

The inlet profile is

$$\mathbf{u}_D = (0.25R(t/0.4)16Y(1-Y)Z(1-Z), 0, 0)^\top, \quad Y = \frac{y}{0.2}, \quad Z = \frac{z}{0.2}. \quad (67)$$

It vanishes at the inlet edges and is imposed by nodal interpolation and finite element Dirichlet lifting.

Eight dipoles are placed outside the channel at

$$\mathbf{x}_i^- = (0.2i, -0.15, 0.1), \quad \mathbf{x}_i^+ = (0.2i, 0.35, 0.1), \quad i = 1, \dots, 4,$$

with directions $\mathbf{d}_i^- = (0, 1, 0)^\top$ and $\mathbf{d}_i^+ = (0, -1, 0)^\top$. We prescribe

$$\Phi_a(\mathbf{x}, t) = \sum_{i=1}^4 a_i(t) \sum_{\epsilon \in \{-, +\}} \frac{\mathbf{d}_i^\epsilon \cdot (\mathbf{x}_i^\epsilon - \mathbf{x})}{|\mathbf{x}_i^\epsilon - \mathbf{x}|^3}, \quad \mathbf{h}_a = \nabla \Phi_a, \quad (68)$$

$$a_i(t) = C_a A R \left(\frac{t - 0.2}{0.4} \right) \left[0.7 + 0.3 \cos \left(2\pi \left(t - \frac{0.2i}{0.8} \right) \right) \right], \quad C_a = 3 \times 10^{-4}. \quad (69)$$

The dipole positions are fixed; the phase shifts act on their time-dependent strengths. The magnetic forcing has period one after the ramp. We take $A = 1$ for the magnetic case and $A = 0$ for a zero-field control with the same inlet data. In the notation of the model, $\mathbf{H}_e = -\mu_0 \mathbf{h}_a$. Since all sources lie outside the fluid, Φ_a is harmonic there. Consequently, the magnetic-potential load is

$$(\mathbf{G}_e, \nabla \psi_h), \quad \mathbf{G}_e = \mu_0 \partial_t \mathbf{h}_a + \frac{\mu_0 + \chi_0}{\tau} \mathbf{h}_a.$$

Open-boundary formulation. We impose $\mathbf{u} = \mathbf{0}$ on Γ_w and the gradient-traction condition

$$((\eta + \zeta) \nabla \mathbf{u} - p \mathbf{I}) \mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_o, \quad (70)$$

where p is the physical pressure and the computed pressure slot is $\tilde{p} = p - \mu_0 \mathbf{m} \cdot \mathbf{H}/2$. The outlet fixes the pressure level, so no pressure-mean constraint is added in these open-channel runs. On the entire boundary we prescribe

$$\begin{aligned} \boldsymbol{\omega} &= \mathbf{0}, & \mathbf{m} \cdot \mathbf{n} &= 0, & \mathbf{k} \times \mathbf{n} &= \mathbf{0}, \\ (\mathbf{H} + \mathbf{m}) \cdot \mathbf{n} &= \mathbf{h}_a \cdot \mathbf{n}, & \int_{\Omega_c} \varphi_h \, dx &= 0. \end{aligned} \quad (71)$$

The tangential trace of $\mathbf{z}_h$ is zero on Γ_w and free on the inlet and outlet. The trace of ϖ_h is zero on $\Gamma_i \cup \Gamma_w$ and free on Γ_o . The weak definition of ϑ_h includes the prescribed normal load:

$$(\vartheta_h, v_h) = (\nabla \varphi_h, \nabla v_h) - \langle \mathbf{h}_a \cdot \mathbf{n}, v_h \rangle_{\partial \Omega_c}.$$

The nine volume equations and their five-block coupling are retained. The open-boundary extension includes the term $\frac{\mu_0}{2} (\text{curl } \mathbf{z}_h, \nabla \psi_h)$ in the magnetic-potential equation. To implement (70) with the gradient/curl form of the momentum equation, its residual also contains

$$\begin{aligned} \mathcal{R}_{\Gamma_o}(\mathbf{v}_h) &= -\langle \mathbf{t}_\Delta, \mathbf{v}_h \rangle_{\Gamma_o} + \frac{\rho}{2} \langle (\mathbf{u}_* \cdot \mathbf{n}) \mathbf{u}, \mathbf{v}_h \rangle_{\Gamma_o}, \\ \mathbf{t}_\Delta &= -\zeta (\nabla \mathbf{u})^\top \mathbf{n} + 2\zeta \mathbf{n} \times \boldsymbol{\omega} \\ &\quad + \frac{\mu_0}{2} [(\mathbf{m}_* \cdot \mathbf{H}) \mathbf{n} + \mathbf{n} \times (\mathbf{H} \times \mathbf{m}_*)]. \end{aligned} \quad (72)$$

Here the unstarred fields in the evolution residuals are midpoint values, while stars denote the prescribed nonlinear-startup or extrapolated coefficients. The inlet lifting and the normal load in the ϑ_h relation use endpoint data. These boundary conditions extend the homogeneous setting of the analysis. The closed-boundary energy estimates are not asserted for this driven open-boundary problem.

Discretization and results. We use the second-order finite element spaces of Section 6.1, $\Delta t = 0.005$, and $T = 2$. The initial step solves the nonlinear midpoint equations; subsequent steps use the specified extrapolation and Algorithm 3, including the fifth-block feedback. Assembly uses quadrature degree eight. The Schur solves use relative and absolute tolerances 10^{-11} and 10^{-16} ; the nonlinear startup uses relative tolerance 10^{-9} with an absolute residual floor of 10^{-14} . The channel meshes contain $30K^3$ tetrahedra, obtained from a $5K \times K \times K$ Cartesian partition. The magnetic runs use $K = 4, 8, 12$, and the control uses $K = 12$.

Figures 5 to 7 show the $K = 12$ magnetic solution at $t = 2$. The velocity retains a predominantly axial profile, whereas the magnetization has a nonuniform distribution set by the dipole field and its coupled response. The particle angular velocity is appreciable in the sheared portions of the flow; it is distinct from the fluid vorticity $\text{curl } \mathbf{u}_h$.

At $T = 2$, native volume- L^2 differences between the $K = 8$ and $K = 12$ solutions, normalized by the $K = 12$ norm, are

Field	$\mathbf{u}$	$\mathbf{m}$	$\boldsymbol{\omega}$	$\mathbf{H}$
Relative difference (%)	0.13524	3.29718	0.71893	3.14611

These are adjacent-mesh differences, not errors against an exact solution. They were integrated on a common refinement and checked with quadrature degrees four and six. The same-mesh zero-field control has zero magnetization and magnetic field. Its differences from the magnetic run are 0.00725% in velocity and 0.15803% in angular velocity in the same native norm, with the magnetic solution as denominator. These small increments have not been independently established to be mesh-converged. The prescribed inlet fixes the through-flow; this comparison concerns field response under magnetic forcing and does not establish a net magnetic pumping gain.

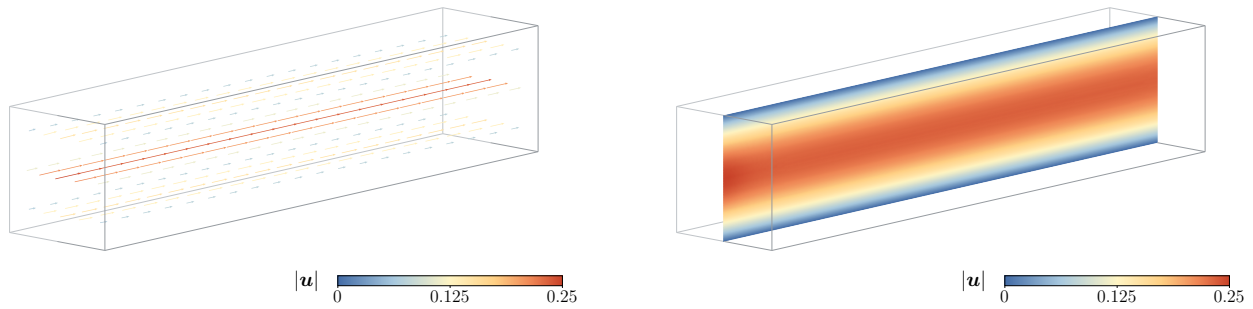


Figure 5: Straight-channel flow at $t = 2$ on $K = 12$: velocity vectors (left) and the velocity magnitude on $z = 0.1$ (right).

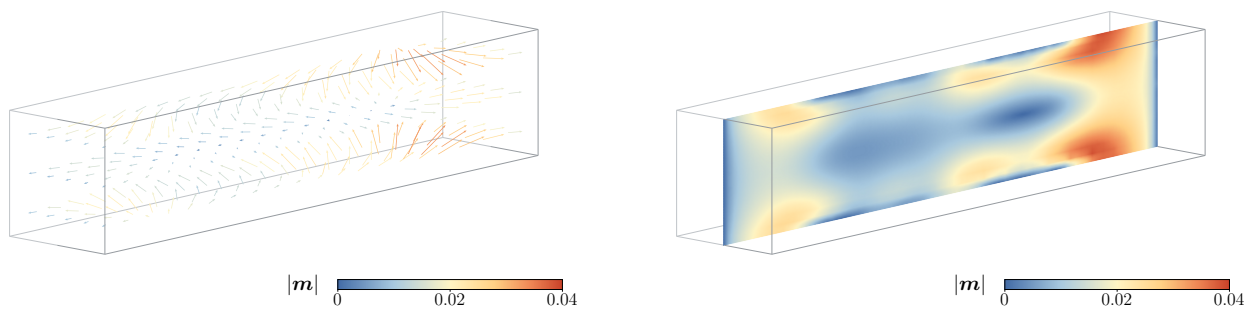


Figure 6: Straight-channel flow at $t = 2$ on $K = 12$: magnetization vectors (left) and the magnetization magnitude on $z = 0.1$ (right).

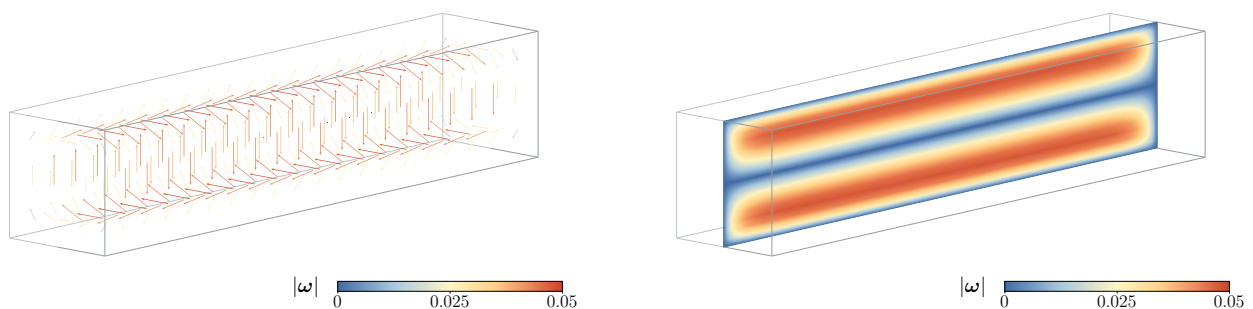


Figure 7: Straight-channel flow at $t = 2$ on $K = 12$: particle angular velocity vectors (left) and their magnitude on $z = 0.1$ (right). The arrows display instantaneous vectors, not particle trajectories.

6.5 Magnetically forced flow over a backward-facing step

To examine the same coupled solver in a channel with a sudden expansion, we consider

$$\Omega_s = (0, 1) \times (0, 0.2) \times (0, 0.2) \setminus ([0, 0.2] \times [0, 0.2] \times [0, 0.1]). \quad (73)$$

The upstream bottom is raised to $z = 0.1$ for $x \leq 0.2$. The fluid volume is 0.036; the inlet $x = 0$, $z > 0.1$ has area 0.02, and the outlet $x = 1$ has area 0.04. The vertical step face and all remaining boundary faces belong to the no-slip wall Γ_w . The geometry and the dipole locations follow the three-dimensional step configuration in [11].

The material parameters, zero initial data, zero body sources, outlet traction, and magnetic and auxiliary boundary conditions are those stated in Section 6.4, with Ω_c replaced by Ω_s . The inlet profile is

$$\mathbf{u}_D = (0.25R(t/0.4)16Y(1-Y)Z_s(1-Z_s), 0, 0)^\top, \quad Y = \frac{y}{0.2}, \quad Z_s = \frac{z-0.1}{0.1}. \quad (74)$$

It is supported on the upper half of the inlet and vanishes on its edges. The inlet and homogeneous wall values are imposed separately, so the polynomial inlet extension is not applied to the vertical step face. After ramp-up, the analytic inlet flux is $1/450$, one half of the straight-channel value because the inlet area is halved.

The applied field retains (68)–(69), with $A = 1$, but uses

$$\mathbf{x}_i^- = (0.2i, -0.2, 0.1), \quad \mathbf{x}_i^+ = (0.2i, 0.4, 0.1), \quad i = 1, \dots, 4, \quad C_a = 7 \times 10^{-4}. \quad (75)$$

The directions remain $\mathbf{d}_i^\pm = \mp(0, 1, 0)^\top$, with the upper sign corresponding to the upper row. Thus the waveform, ramp times, period, and phase speed are unchanged, while the magnetic-source distance and prefactor are explicitly different. In particular, the relaxation time remains $\tau = 0.1$ and the diffusion coefficient remains $\sigma = 0.005$.

Mesh and time stepping. We use the same second-order spaces, nonlinear midpoint startup, subsequent extrapolation, quadrature, and solver tolerances as in the straight channel. The time step is 0.005 and the simulation runs to $T = 2$. A conforming tetrahedral mesh is formed from a $5K \times K \times K$ background partition, with the solid cells removed. Before extracting the fluid mesh, its logical axial coordinate ξ is mapped to x by the continuous piecewise affine function specified by

ξ	0	0.1	0.2	0.6	1
x	0	0.15	0.2	0.4	1.

This grades the mesh near the step; the transverse coordinates remain uniform. The resulting mesh has $27K^3$ tetrahedra. The following visualizations use $K = 4$, namely 1728 tetrahedra.

Field visualization and interpretation. Figures 8 to 10 use the actual saved time layer $t = 0.715$ (step 143), rather than a temporally interpolated state. All three fields use this same time. The velocity spreads into the larger downstream cross-section after the step. The magnetization exhibits spatially varying direction and magnitude, while the angular-velocity distribution is influenced by the shear near the step and the channel walls. The displayed angular velocity is the particle spin.

The right panels show the full three-component magnitude on $y = 0.1$. The magnetization arrows are sampled on that same plane; the velocity and angular-velocity arrows use sparse three-dimensional locations. Solid-region samples are excluded. The linear arrow scaling and the color scales are fixed over the stored time interval. Point values come from the original finite element fields without RT vertex averaging; interpolation in the filled color surface is used only for display, not for evaluating finite element norms. The color limits for $\mathbf{u}$, $\mathbf{m}$, and $\boldsymbol{\omega}$ are 0.25, 0.04, and 0.04, respectively.

The 400-step $K = 4$ trajectory passed the stated algebraic and boundary checks: its maximum recomputed relative residual was 2.03×10^{-10} , and its largest recorded relative inlet–outlet flux imbalance was 1.12×10^{-14} . A full-interval $K = 4$ comparison with $\Delta t = 0.0025$ gave endpoint relative native volume- L^2 differences below 2.9×10^{-5} for $\mathbf{u}$, $\mathbf{m}$, $\boldsymbol{\omega}$, and $\mathbf{H}$. These checks assess the implementation and time-step sensitivity. At the displayed spatial resolution, the figures are a qualitative illustration; quantitative mesh independence of the step-flow observables remains to be assessed. In particular, no resolved reattachment length, corner extremum, or magnetic pumping efficiency is inferred from these plots. The nonconvex geometry and open boundary conditions also lie outside the homogeneous convex-domain setting used in the analysis.

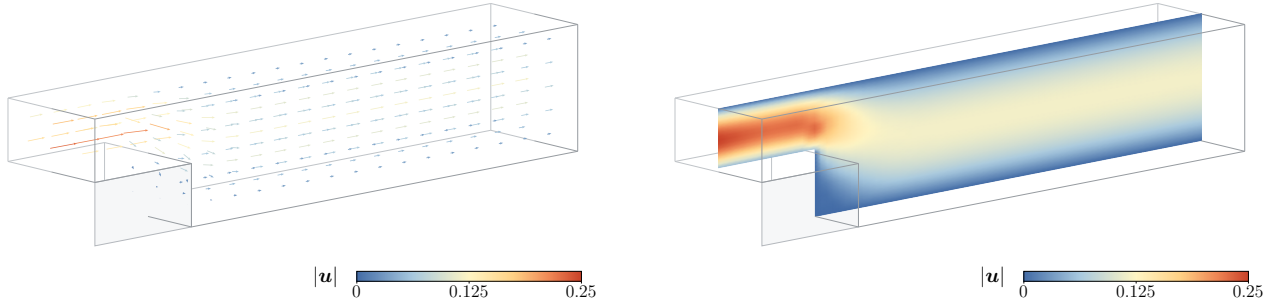


Figure 8: Backward-facing-step flow at $t = 0.715$ on $K = 4$: velocity vectors (left) and the velocity magnitude on $y = 0.1$ (right). The unfilled upstream region is the solid step.

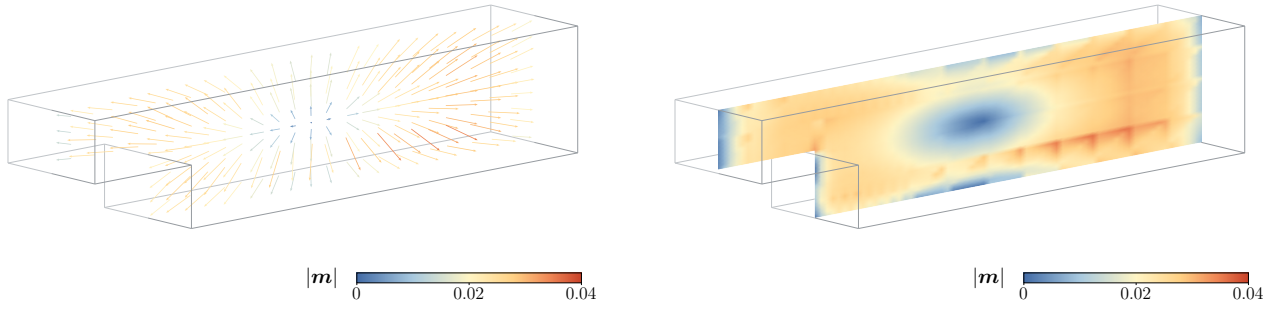


Figure 9: Backward-facing-step flow at $t = 0.715$ on $K = 4$: magnetization vectors (left) and their full magnitude (right), both sampled on $y = 0.1$.

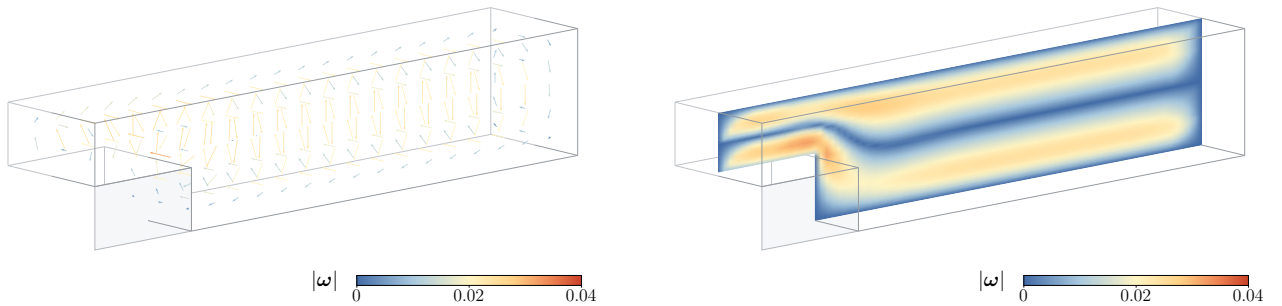


Figure 10: Backward-facing-step flow at $t = 0.715$ on $K = 4$: particle angular velocity vectors (left) and their magnitude on $y = 0.1$ (right).

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